Thinking causally

Estimands, DAGs, and model specification

Sebastian Ramirez-Ruiz
European University Institute (EUI)
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I am Sebastian Ramirez-Ruiz, or just Sebastian.

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Both questions come **before** you choose a model.

This session

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Causation and causal reasoning, counterfactuals, potential outcomes, and the biases of naive comparisons

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Practice

Some group discussion today; DAGs and simulations in R in Lab 3 on Monday

Why causation matters

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Causation: a poor man's working definition

- a.k.a. causality, cause and effect
- The idea that one variable [event, process, object; the cause] leads to another variable [event, process, object, outcome; the effect]

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Causal reasoning

- The process of studying and identifying **causality** (*ruling out non-causal sources of association*)
- **Causal inference** is one way of doing causal reasoning

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> **Side note: what can be a cause?** The **manipulationist** camp holds there is **no causation without manipulation** (Holland 1986): only things we could, in principle, intervene on are causes, so "the effect of race" or "the effect of sex" is *not a thing* for them. The **difference-making** (or structural) camp counts anything that makes a difference to the outcome. Others break attributes into manipulable parts (Sen & Wasow 2016: race as a "bundle of sticks"). That is why we start with a loose, working definition and get precise once we define potential outcomes.

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- you opened social media and saw **yet another new estimator.** What to do?

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- you opened social media and saw **yet another new estimator.** What to do?
- **ultimately, you think some X might affect Y and want to find out**

Does going to the opera make you live longer?

NYT Health
@NYTHealth

Want to live longer? Try going to the opera. Researchers in Britain have found that people who reported going to a museum or concert even once a year lived longer than those who didn't.



Another Benefit to Going to Museums? You May Live Longer (Published 2019)
Researchers in Britain found that people who go to museums, the theater and the opera were less likely to die in the study period than those who didn't.
nytimes.com

3:19 PM · Dec 22, 2019 · SocialFlow

318 Retweets **1,705** Quote Tweets **1,239** Likes

Does going to the opera make you live longer?



Researchers in London who followed thousands of people 50 and older over a 14-year period discovered that those who went to a museum or attended a concert just once or twice a year were 14 percent less likely to die during that period than those who didn't.

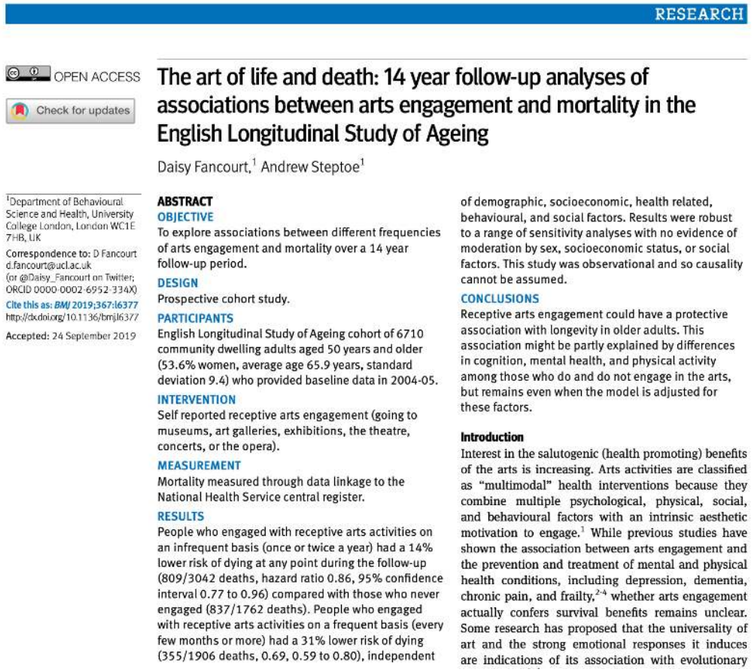
The chances of living longer only went up the more frequently people engaged with the arts, according to the study, which was published this month in The BMJ, formerly The British Medical Journal. People who went to a museum or the theater once a month or even every few months had a 31 percent reduced risk of dying in that period, according to the study.

The study controlled for socioeconomic factors like a participant's income, education level and mobility, said Andrew Steptoe, a co-author of the study and the head of University College London's research department of behavioral science and health.

Source: Cramer, NYTimes (2019)

Does going to the opera make you live longer?





Source: Fancourt, Steptoe / BMJ (2019)

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Does going to the opera make you live longer?



Different sources of association between X and Y

Does going to the opera make you live longer?



Different sources of association between X and Y

1. X may have caused Y.



Does going to the opera make you live longer?



Different sources of association between X and Y

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2. Y may have caused X.

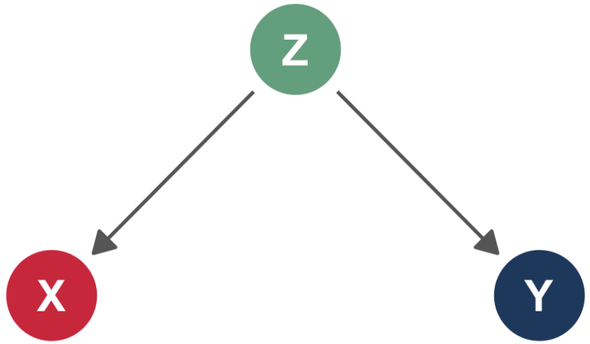


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Different sources of association between X and Y

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2. Y may have caused X.
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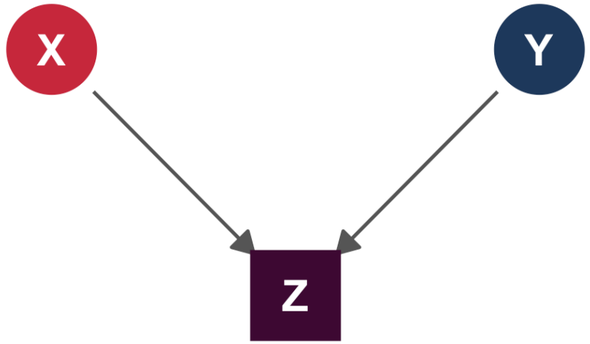


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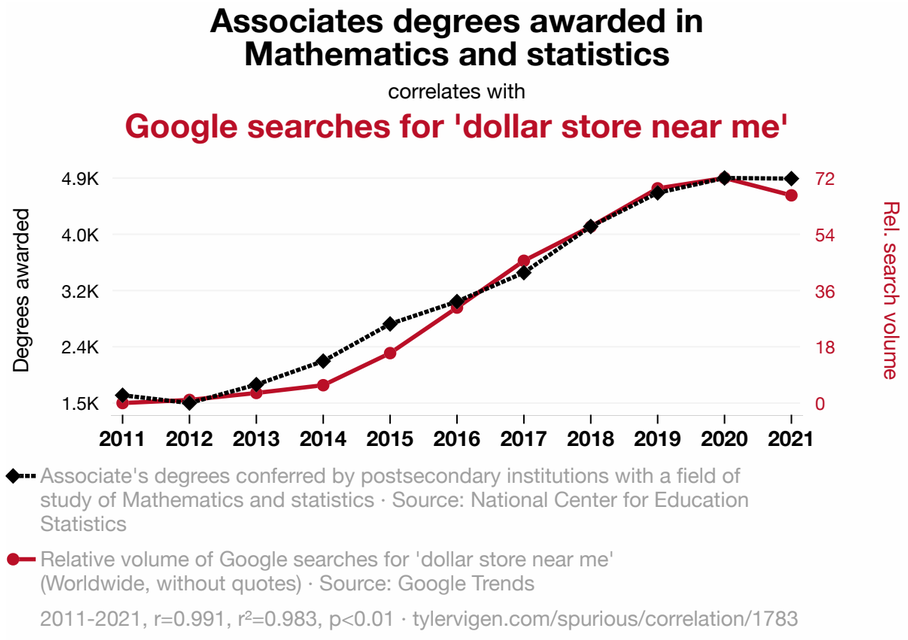
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2. Y may have caused X.
3. X and Y may share one or more common causes.
4. The association may have been induced by conditioning on a common effect of X and Y.
5. Chance.

Empirical regularities, causality, and the enterprise of writing papers… (?)



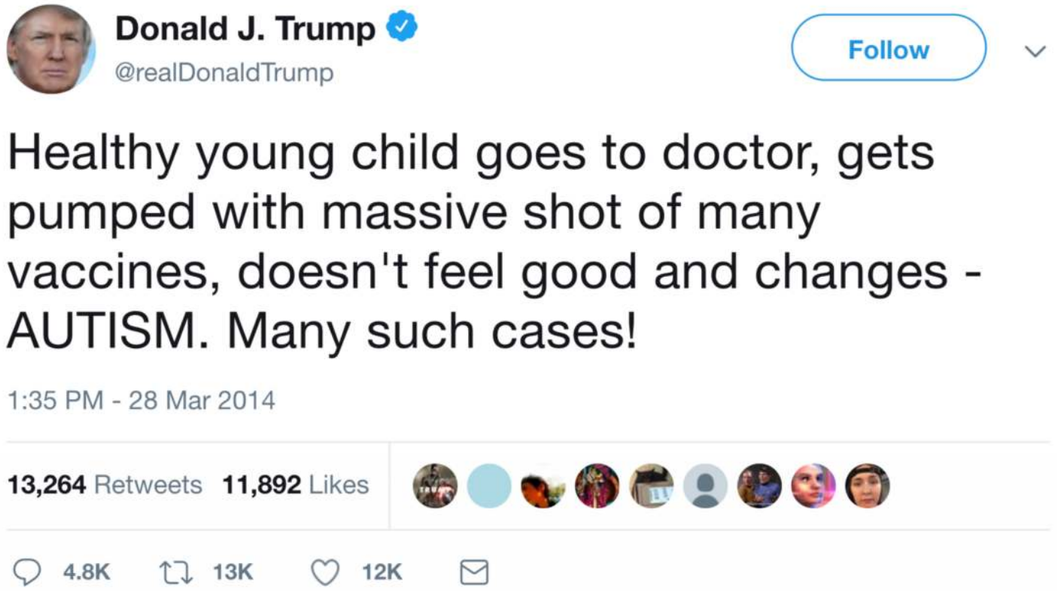
Source: Tyler Vigen | AI paper

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Why causality is important



From associations to causes

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So, how do we figure out correlation?

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So, how do we figure out correlation?

Math and statistics.

From associations to causes

So, how do we figure out correlation?

Math and statistics.

And how do we figure out causation?

From associations to causes

So, how do we figure out correlation?

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And how do we figure out causation?

Philosophy. Imagination. Theory. Counterfactual thinking. No (*some*) math.

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- Our understanding of causality is **probabilistic** rather than deterministic: X is a cause of Y if X is a cause of Y for some subjects in a population (or if changes in X lead to changes in the probability distribution in Y)

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- Our understanding of causality is **probabilistic** rather than deterministic: X is a cause of Y if X is a cause of Y for some subjects in a population (or if changes in X lead to changes in the probability distribution in Y)
- Most of the time, we are interested in the **existence and magnitude of a causal effect.** Interest in the **mechanism** will come later (for now)

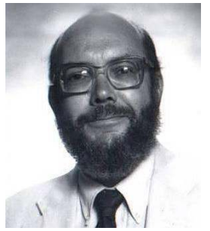
Counterfactual causality

Thinking in terms of counterfactuals

The Neyman-Rubin causal model

- It is a formal approach to do causal inference that fueled the **potential outcomes framework (POF)**
- Up until today, it's the mainstream framework to (statistically) talk about causation







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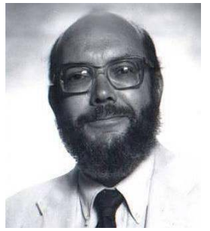
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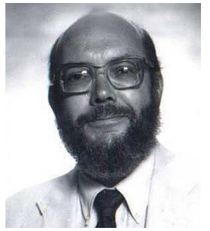
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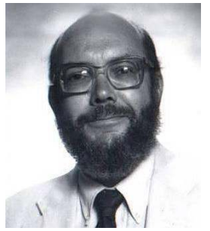
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- The idea is to think about the counterfactual outcome that would have been observed had the treatment been different



Running example: roommates and the contact hypothesis

- **Context:** Prejudice between ethnic groups is persistent and politically consequential. Allport's **contact hypothesis** holds that, under the right conditions, interpersonal contact with members of an out-group reduces prejudice.
- **Intervention:** First-year students share dorm rooms. Some live with a roommate from a different ethnic group (an **out-group roommate**), others with someone from their own group.

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Research question: Does living with an out-group roommate reduce a student's **prejudice**?

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- Evidently, each subject can only be observed in one treatment state *at any point in time*
- Therefore, the ITE is a non-observable counterfactual (what Holland, 1986, labels the **fundamental problem of causal inference**)
- Note that even successive exposure of the same subjects to alternative causal states does not allow for the observation of the ITE — **Why?**

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$$
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Y_i &= Y_i(1) \text{ if } D_i = 1 \\
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Group	Under treatment $(D = 1)$	Under control $(D = 0)$
Treated $(D = 1)$	Observable as $Y_i(1)$	Counterfactual $Y_i(0)$
Untreated $(D = 0)$	Counterfactual $Y_i(1)$	Observable as $Y_i(0)$

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Why is the causal effect of D on Y not simply the difference between the upper-left and lower-right cell in the table?

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- **NATE will correspond to ATE only under very specific conditions** regarding the process by which subjects are assigned to causal states. Think of:

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  - A subject decides itself whether to enter one causal state or another (students choose their roommates)

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  - Some combination thereof

Eight students: the world with full information

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6	5	−1	0
2	4	2	−2	1
3	4	4	0	0
4	6	7	1	0
5	3	1	−2	1
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Eight first-year students. We imagine we can see both potential outcomes (prejudice, 0–10).

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Average of the last column:

$$\text{ATE} = \frac{-4}{8} = -0.5$$

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…and what we actually observe

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1	6			0
2		2		1
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The naive comparison says contact cuts prejudice by almost 4 points. The true average effect is **half a point**.

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- Contact "works" for those who sought it out, and **not for the rest**. Which one is your estimand?

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- Using the above expressions, we can further **decompose total bias into its component parts**

Defining biases

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\begin{aligned}
\text{Total bias} &= \text{NATE} - \text{ATE} \\
&= E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] - \{\pi \cdot \text{ATT} + (1-\pi) \cdot \text{ATC}\} && \text{definitions} \\
&= \big(E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]\big) + \big(E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]\big) - \{\cdots\} && \pm E[Y_i(0) \mid D_i = 1] \\
&= \text{ATT} + \big(E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]\big) - \{\text{ATT} - (1-\pi)(\text{ATT} - \text{ATC})\} && \pi = 1 - (1-\pi) \\
&= \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{baseline bias } = 3 - 5.6 \,=\, -2.6} + \underbrace{(1-\pi)\big(\text{ATT} - \text{ATC}\big)}_{\text{differential treatment effect bias } = \frac{5}{8}(-1.33) \,=\, -0.83} && \text{ATTs cancel}
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- where **baseline bias** is the difference in the average outcome *in the absence of treatment* between those under treatment and those under control: students who chose out-group roommates were **less prejudiced to begin with**
- and **differential treatment effect bias** is the difference in the average treatment effect between treated and control groups, weighted by the share in the control state: contact **helps more** those who sought it
- $\therefore$ Total bias = **baseline bias** + **differential treatment effect bias** $= -2.6 - 0.83 = -3.43$

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What we can estimate with “realized” outcomes = True effect

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If SUTVA fails, the **estimand itself** is not defined, before any data problem.

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But **which X?** That is what DAGs answer.

What is your estimand?

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```
lm(prejudice ~ outgroup_roommate + openness + outgroup_friends, data = dorms)
```

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Same (*code*) test, three questions. **The (*code*) test does not tell you which one you answered.**

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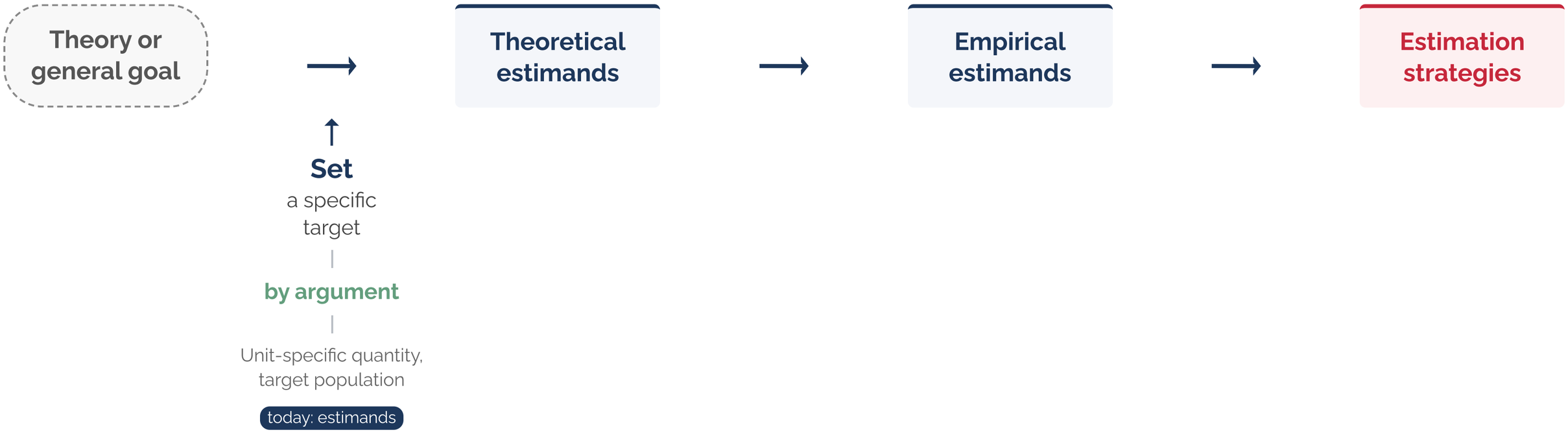
Say what.

Set, link, learn: from theory to evidence



Lundberg, Johnson & Stewart (2021), "What is your estimand?", *American Sociological Review* 86(3)

Set, link, learn: from theory to evidence



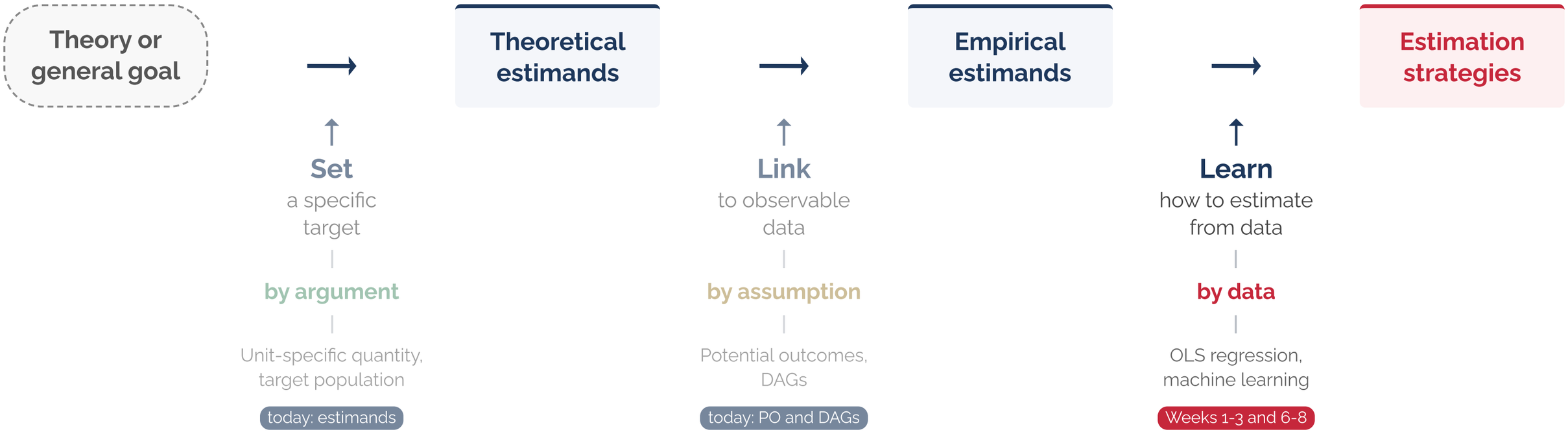
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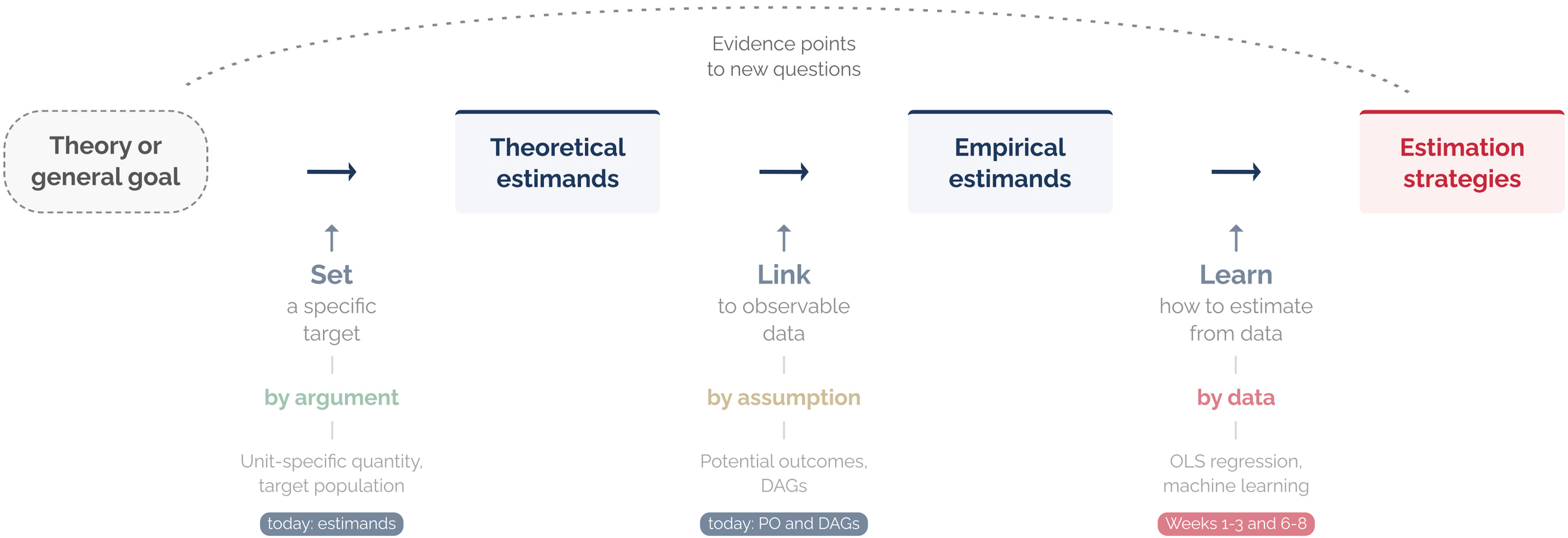
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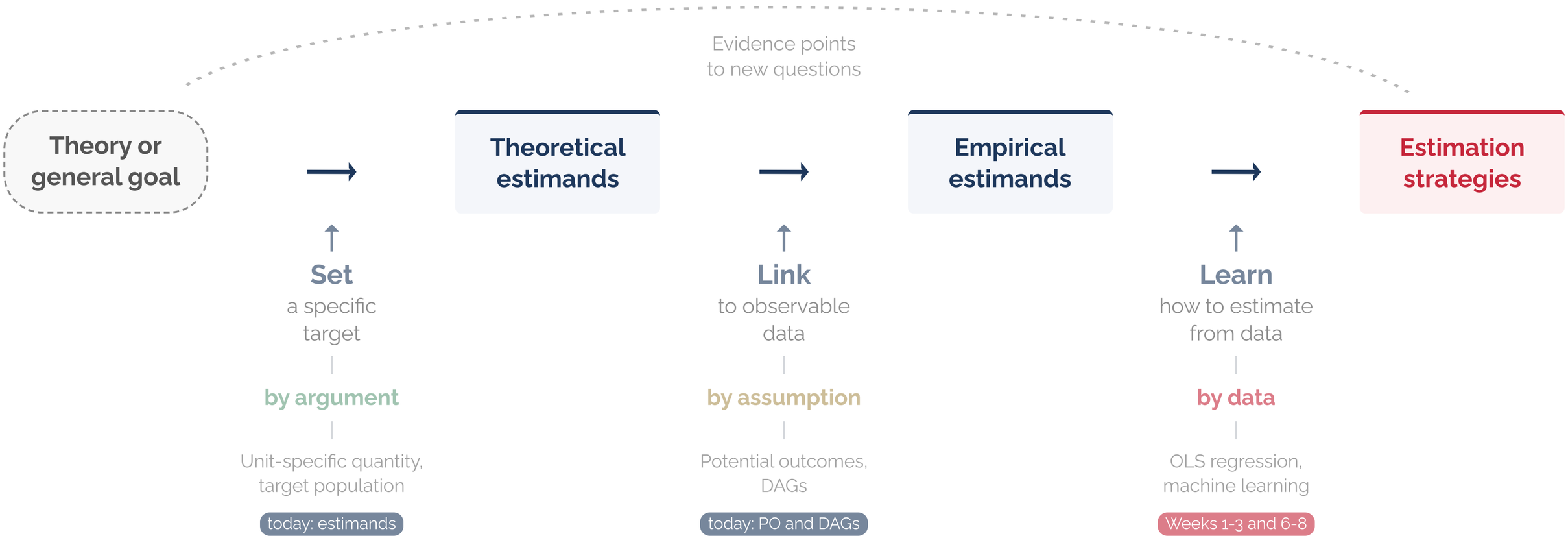
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Most papers start with the estimation strategy. **Lundberg, Johnson & Stewart** ask us to start by **setting** the target.

Lundberg, Johnson & Stewart (2021), "What is your estimand?", *American Sociological Review* 86(3)

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Notice what is missing: **no regression, no β, no control variables.**

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- No counterfactual, no DAG, no "controls" needed.
- Perfectly legitimate, if you **say so** and do not slip into causal language.
- The trouble starts when a descriptive estimate is **read as a causal one**.

The target population is part of the question

Boisjoly, Duncan, Kremer, Levy & Eccles (2006), *AER* 96(5) · Behaghel, Crépon & Le Barbanchon (2015), *AEJ: Applied* 7(3)

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A perfectly identified effect can still answer **the wrong population's question**. We come back to this as **transportability** in Week 4.

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- The **Learn** step, estimation (OLS, matching, weighting…), comes **last**.

Your turn ✏️ (~3 min, with the person next to you)

Pick **one** question and write down its **unit-specific quantity** and **target population**.
Is it descriptive or causal?

1. Does living with an out-group roommate reduce prejudice?
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Would a regression of Y on the treatment plus controls give you that quantity? Why (not)?

Break

10 minutes

DAGs: drawing your assumptions

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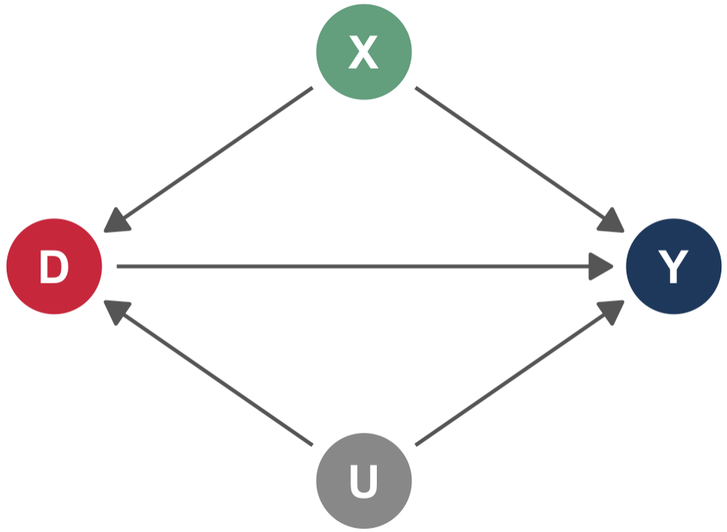
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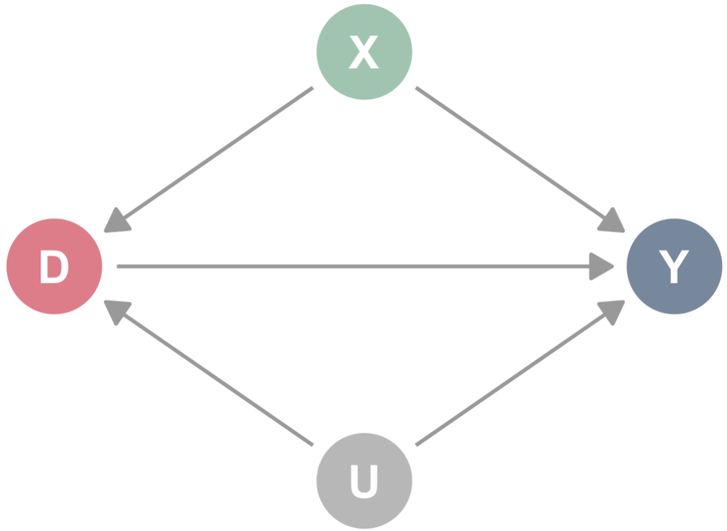
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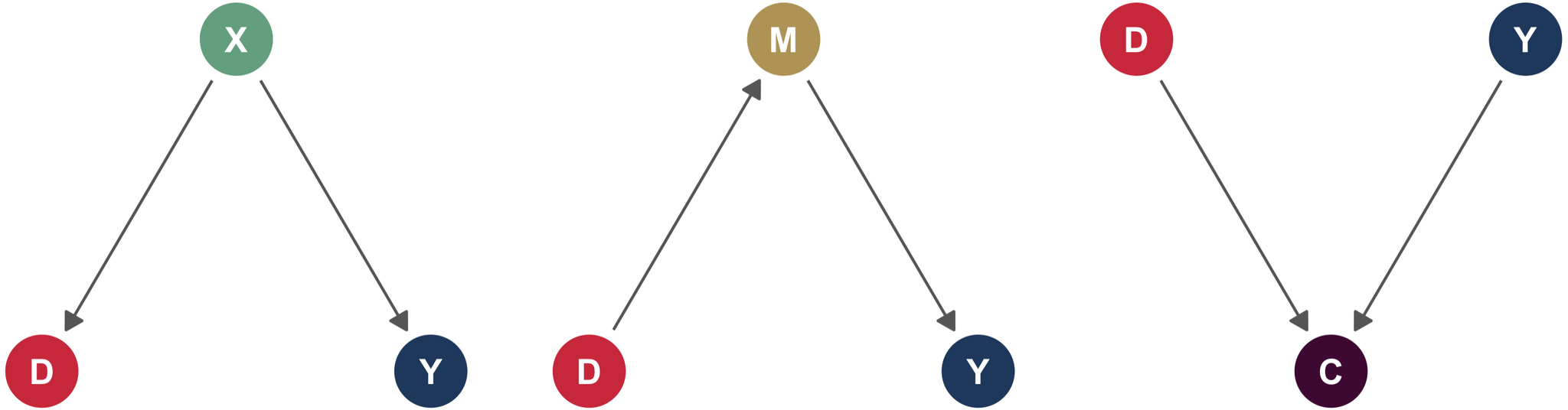
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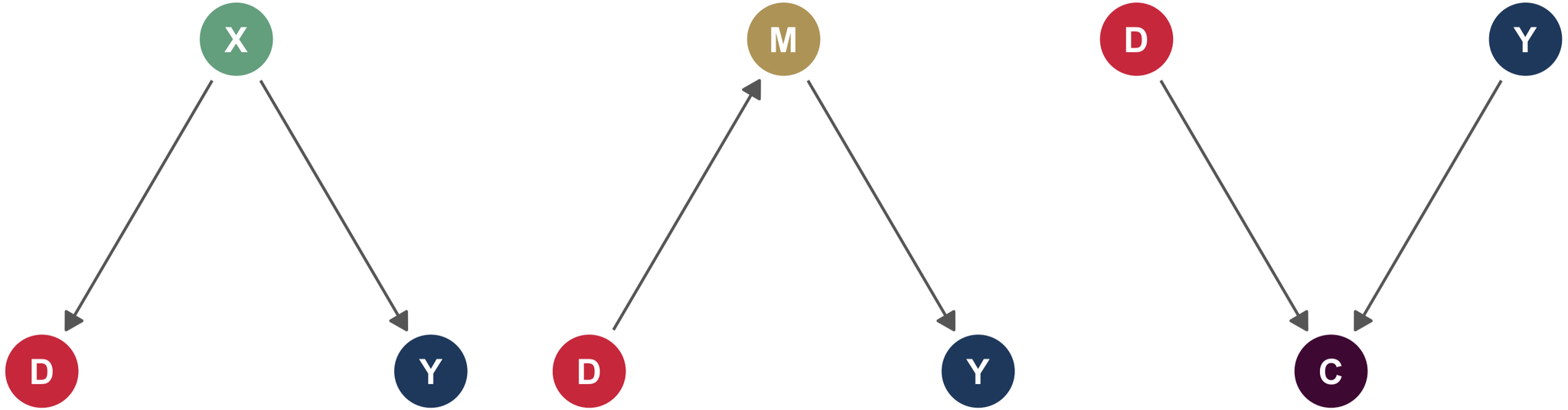
Two languages for the same problem, with different strengths: DAGs make complex assumptions **visible**; potential outcomes tie them to **assignment, heterogeneity, and estimation** (Imbens 2020).

Three building blocks: confounders, mediators, colliders



In the DAG literature, these three structures are also called a *fork*, a *chain*, and an *inverted fork* (or collider).

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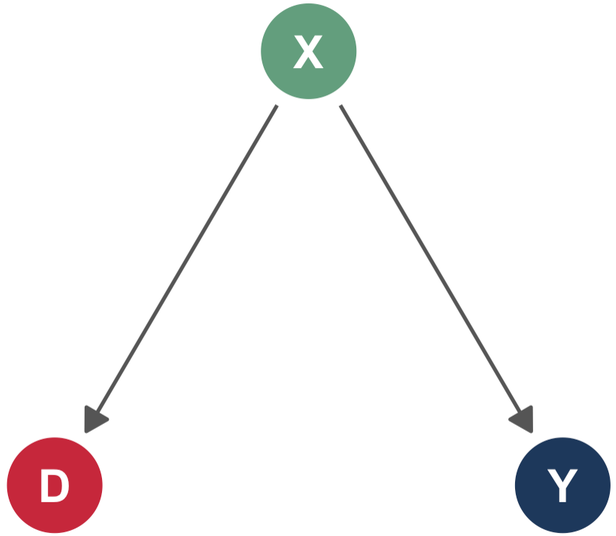


X is a **confounder**

D ← X → Y

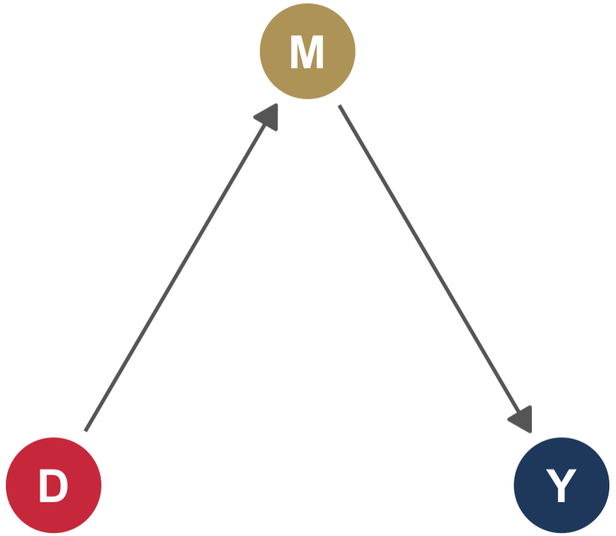
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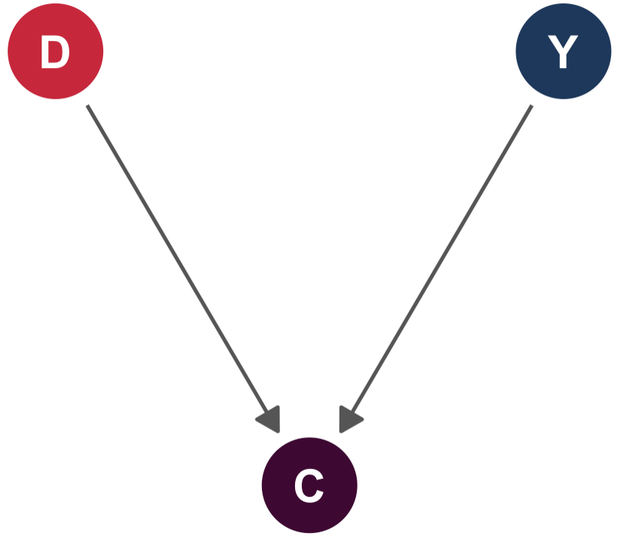
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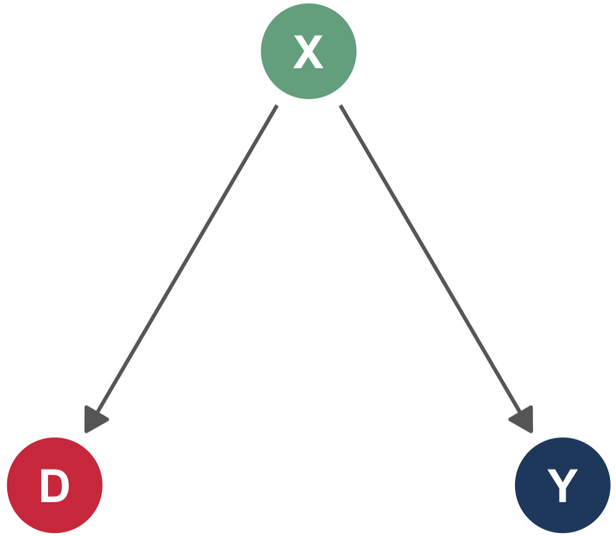
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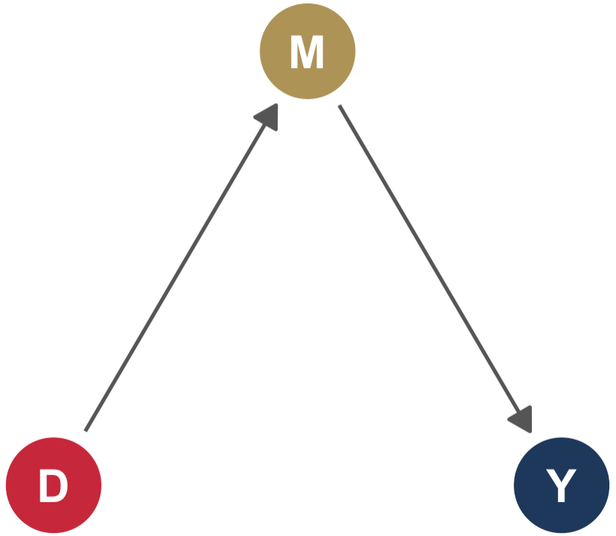
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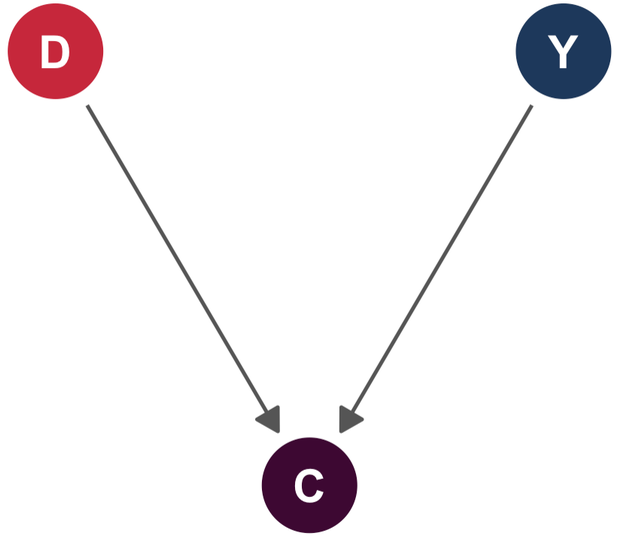
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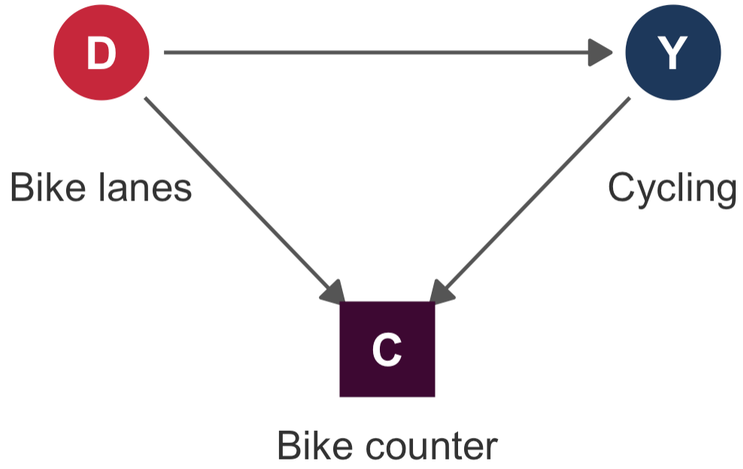
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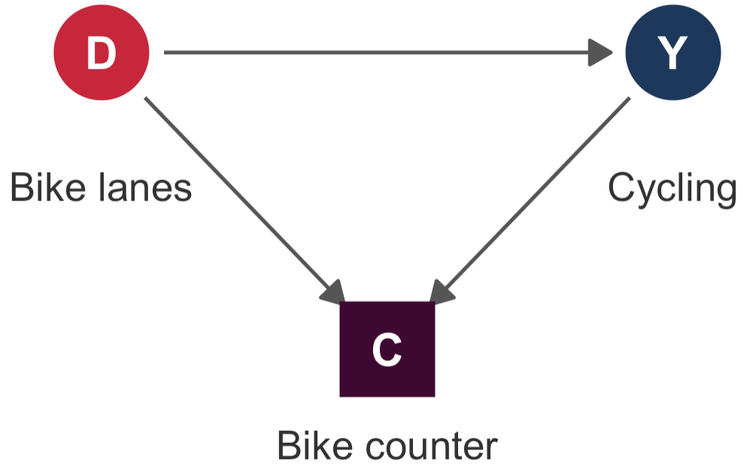


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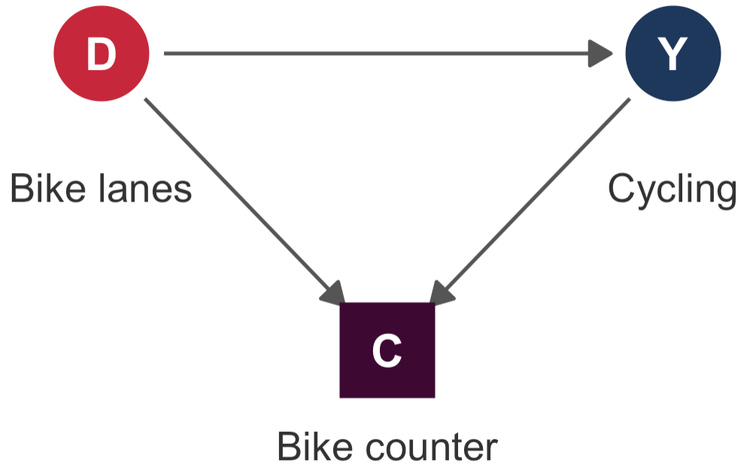


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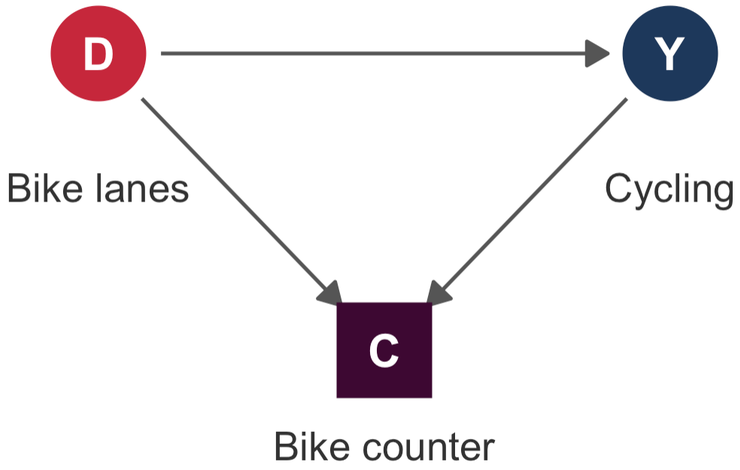


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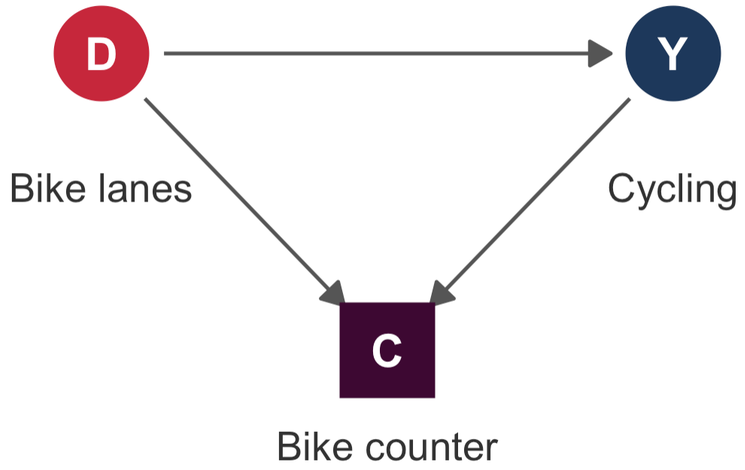


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- So no-lane districts look like heavy-cycling places: lanes and cycling become **negatively associated** among counter districts.

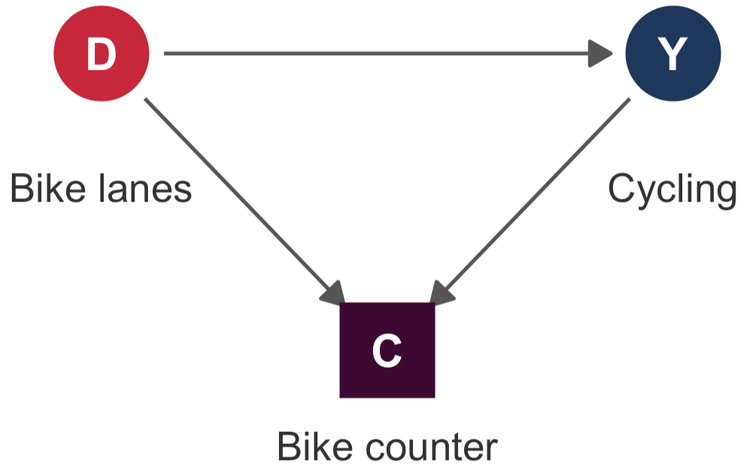


Conditioning on a collider: the bike-counter problem

"Conditioning" = controlling in a regression, stratifying, matching, **or selecting the sample**.

Fair warning: I am very much into cycling (and a bit unidimensional), so expect bike stuff to keep popping up in this course.

- Do **protected bike lanes** increase **cycling**? We only have data from districts with automatic **bike counters**.
- We have reason to believe that cities install counters where lanes exist **or** where cycling is already high: the counter is a **collider**.
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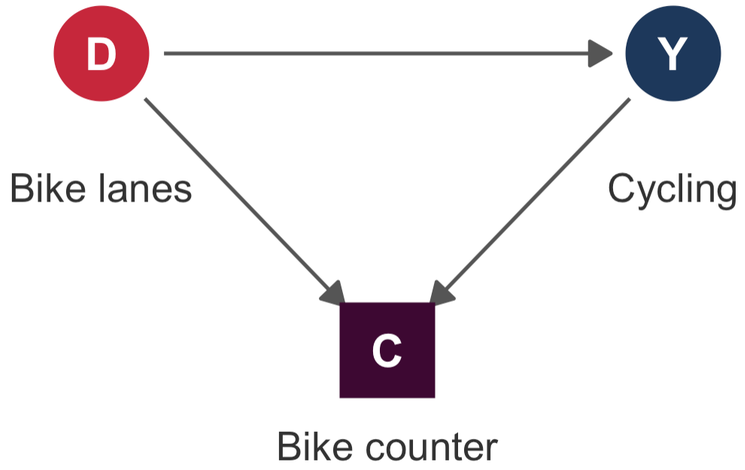


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- More counter districts do not help: the bias comes from **who is in the data**, not from sample size.



The backdoor criterion

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A set of variables Z identifies the effect of D on Y if

1. no variable in Z is a **descendant** of D, and
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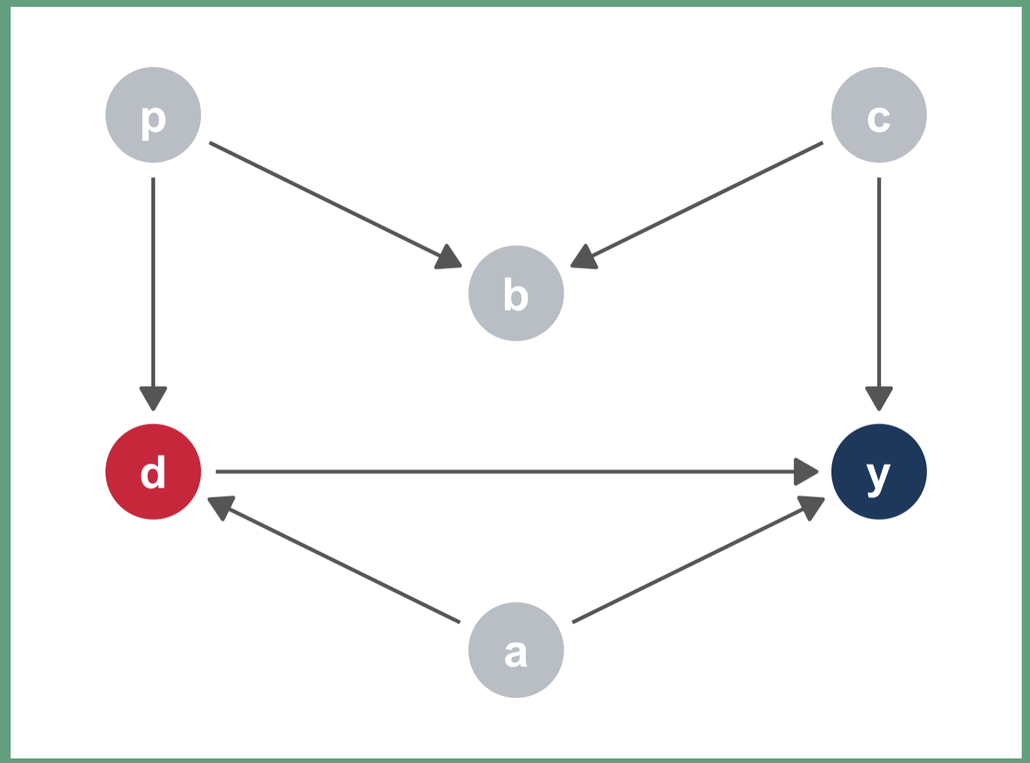
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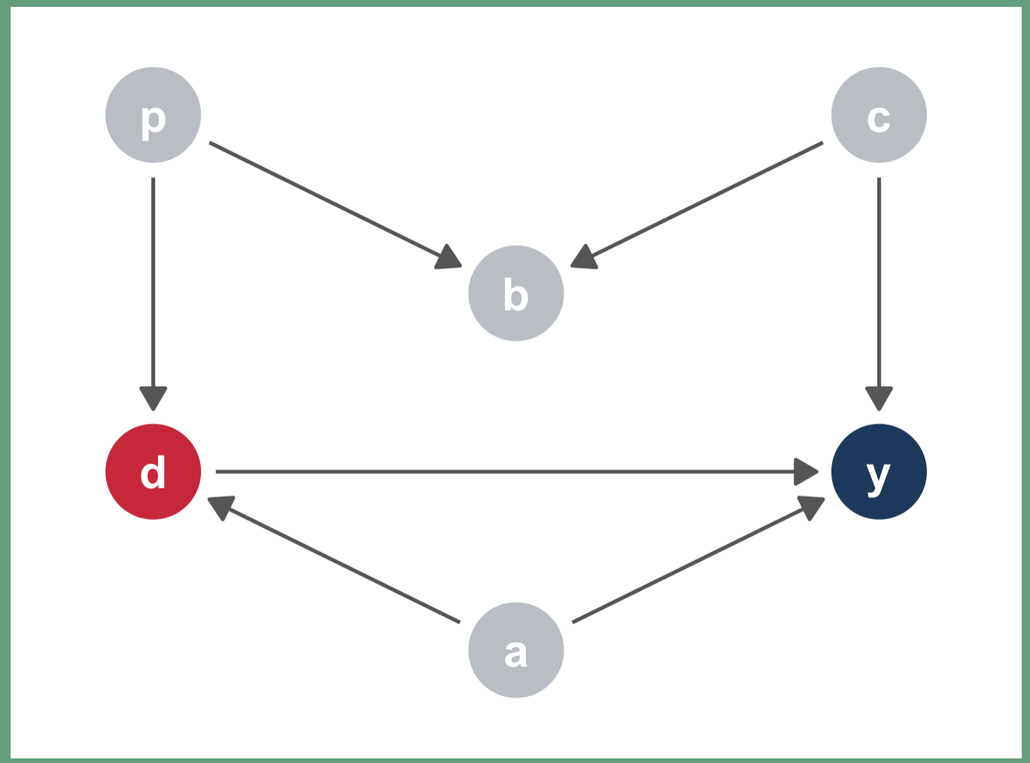
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2. List all paths from D to Y.
3. Mark the backdoor paths, and whether each is open or closed.
4. Find a set that closes the open ones **without** opening new ones (Rule 2!).

Let's do this together 🔍



1. Identify the **causal** and **non-causal** paths from d to y.
2. Are the non-causal paths **naturally closed** (by a collider)?
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A colleague controls for b "to be safe". What else must they add?

Good and bad controls

More controls is not safer

Cinelli, Forney & Pearl (2024), "A crash course in good and bad controls", *Sociological Methods & Research* 53(3)

More controls is not safer

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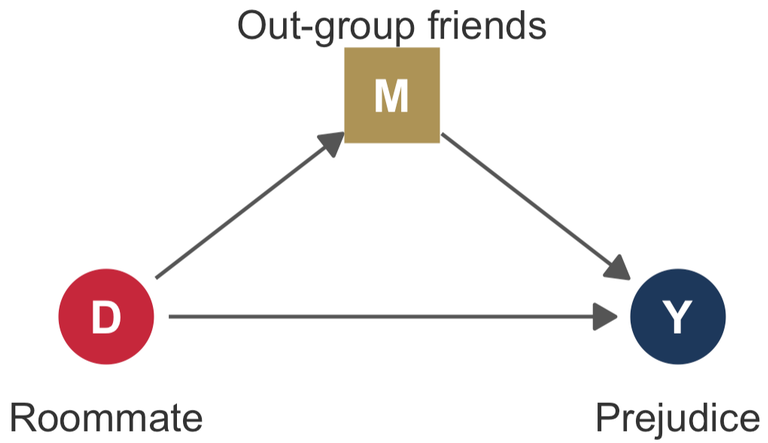
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Bad: creates bias

Mediators, colliders and their descendants, descendants of the treatment, and "M-bias" pre-treatment colliders

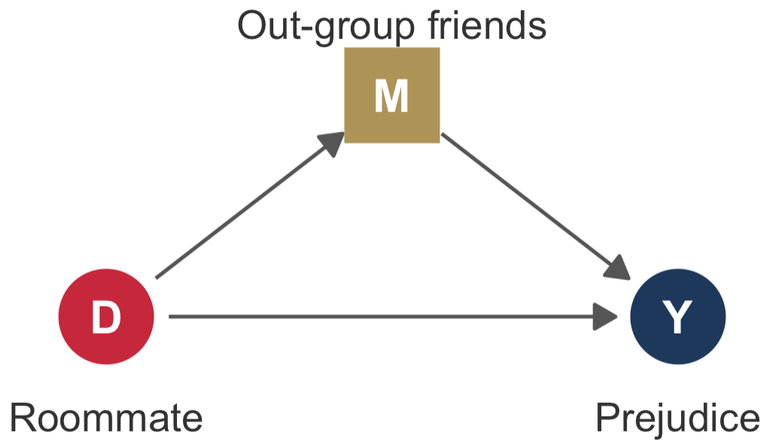
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Bad control 1: controlling away the effect



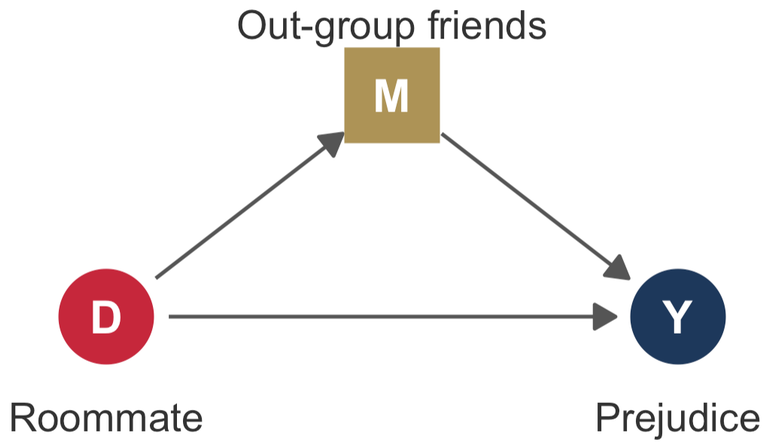
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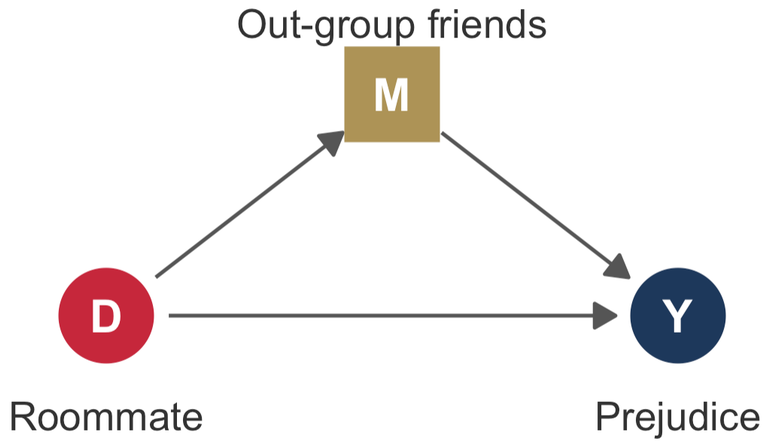
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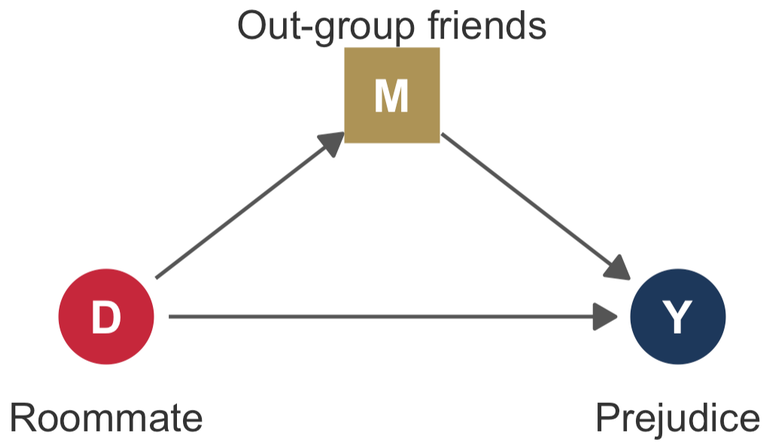
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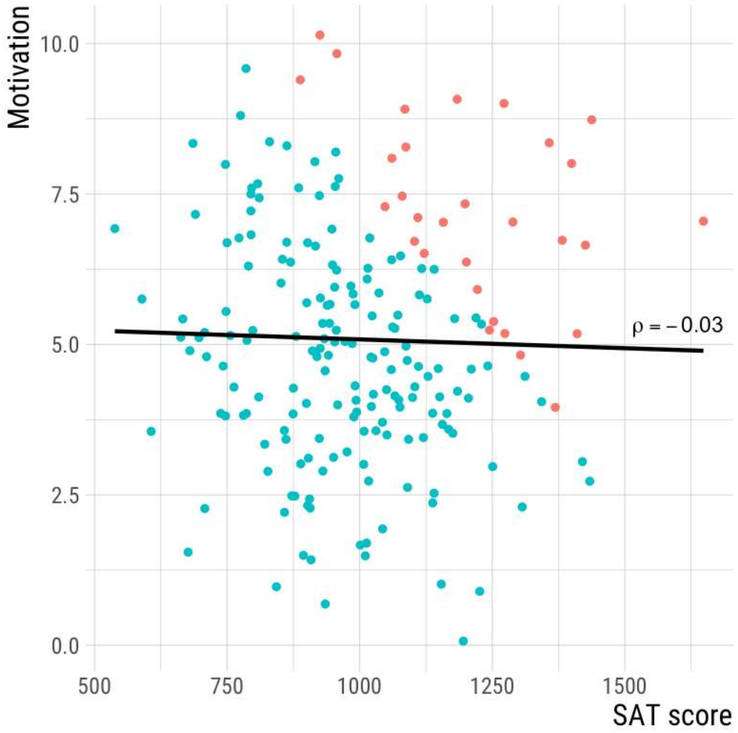


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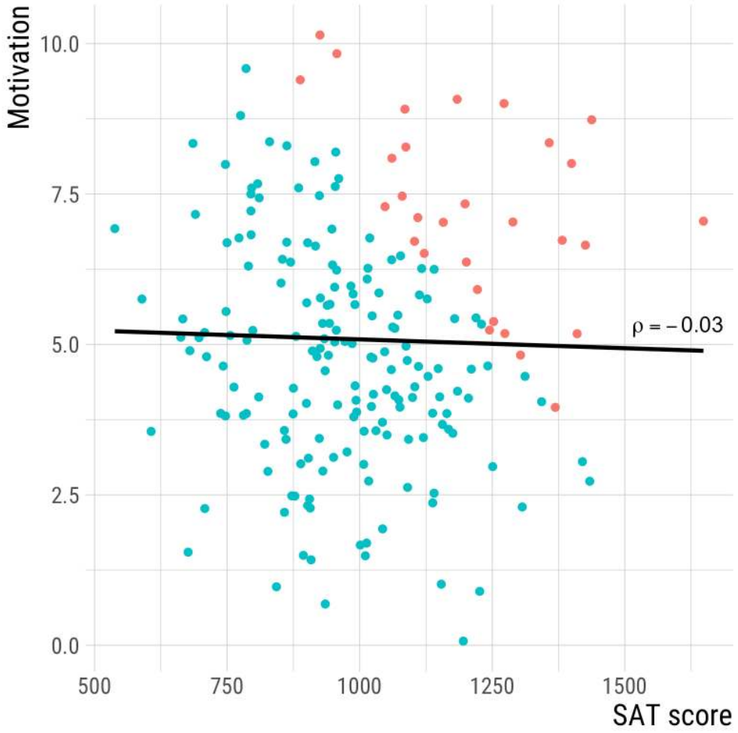
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- Same problem with any **post-treatment** variable, including in experiments: controlling for it, dropping people on it, or subsetting on it (Montgomery, Nyhan & Torres 2018).

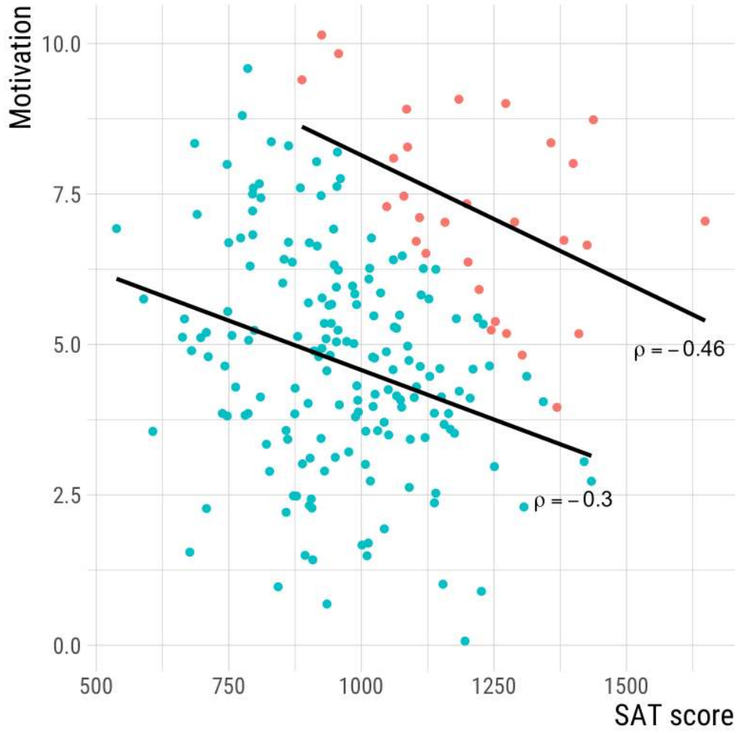


Bad control 2: colliders manufacture correlation

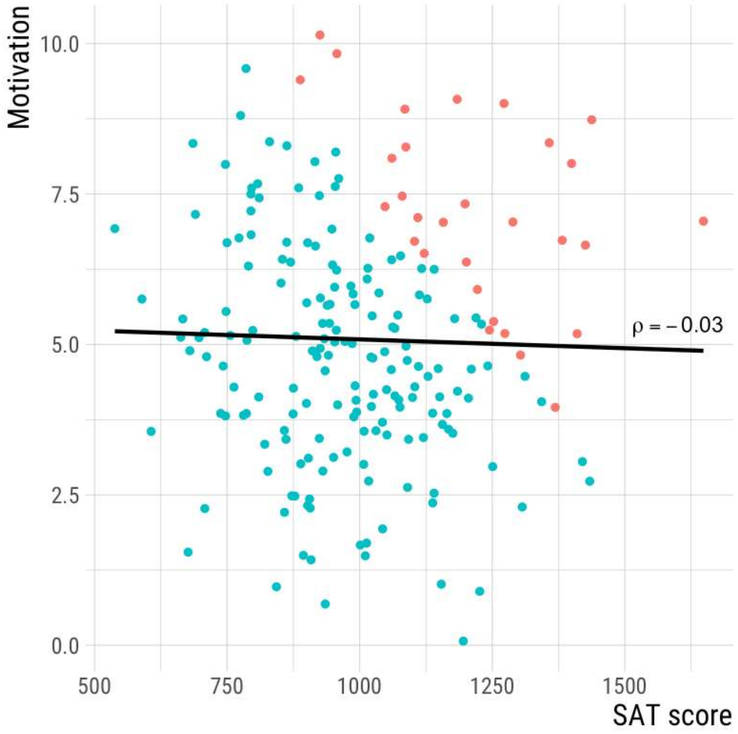


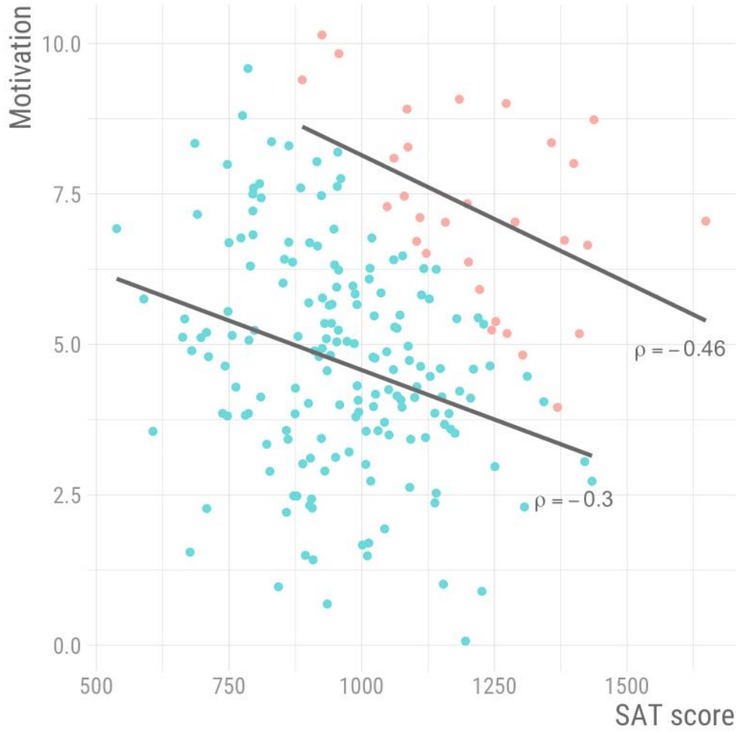
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SAT score and motivation are independent, but both get you **admitted**. **Among admitted students**, they look negatively related.

In the lab on Monday, you will simulate both mistakes:

Skip a confounder: shoe size "predicts" salary, until you control for **sex**.

Control for a collider: beauty and talent look negatively related, once you condition on **celebrity**.

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The things we study are not. Contact and prejudice, bike lanes and cycling, race and police force.

Ask yourself how obvious the **right model** would be there.

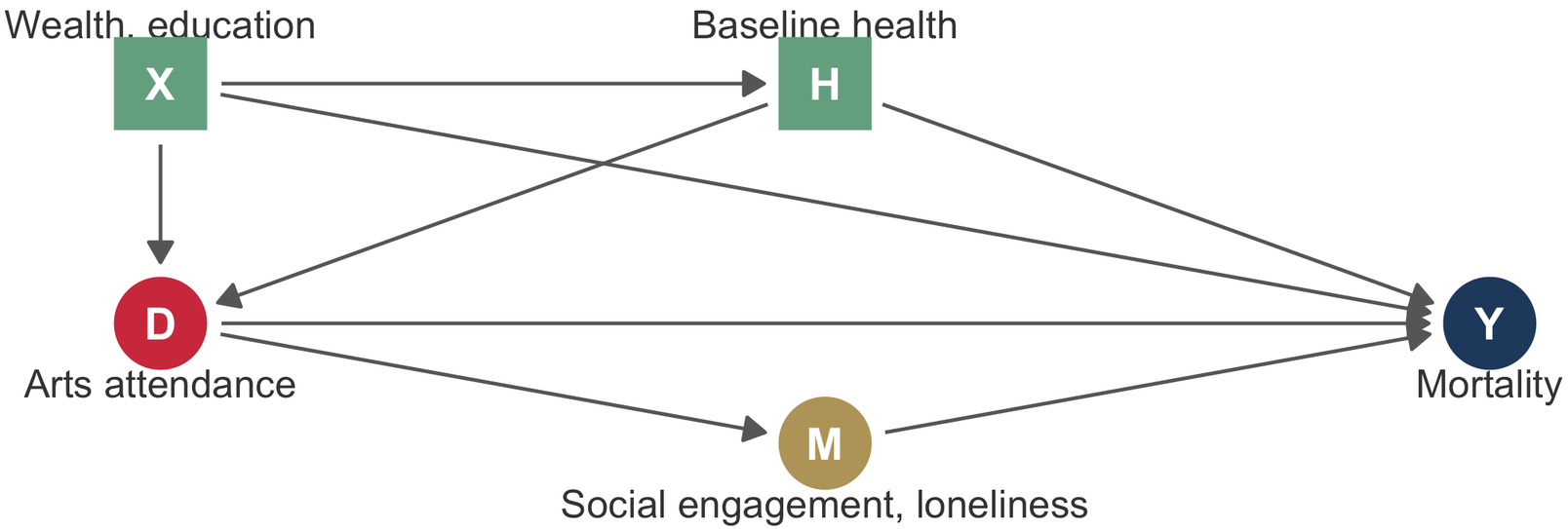
Opera, revisited: what did the authors control for?



> *"We adjusted models for **demographic variables** (age, sex, marital status, ethnicity, educational qualifications, wealth, employment status, and occupational status); **health related variables** (eyesight, hearing, depressive symptoms, other psychiatric conditions, diagnosis of cancer, lung disease or cardiovascular disease, history of any other long-term condition, smoking, alcohol consumption, sedentary behaviours, mobility, problems in undertaking activities of daily living, osteoporosis, and cognition); and **social covariates** (loneliness, number of close friends, living alone, frequency of civic engagement, frequency of social engagement, and whether participants had a hobby or pastime)."*

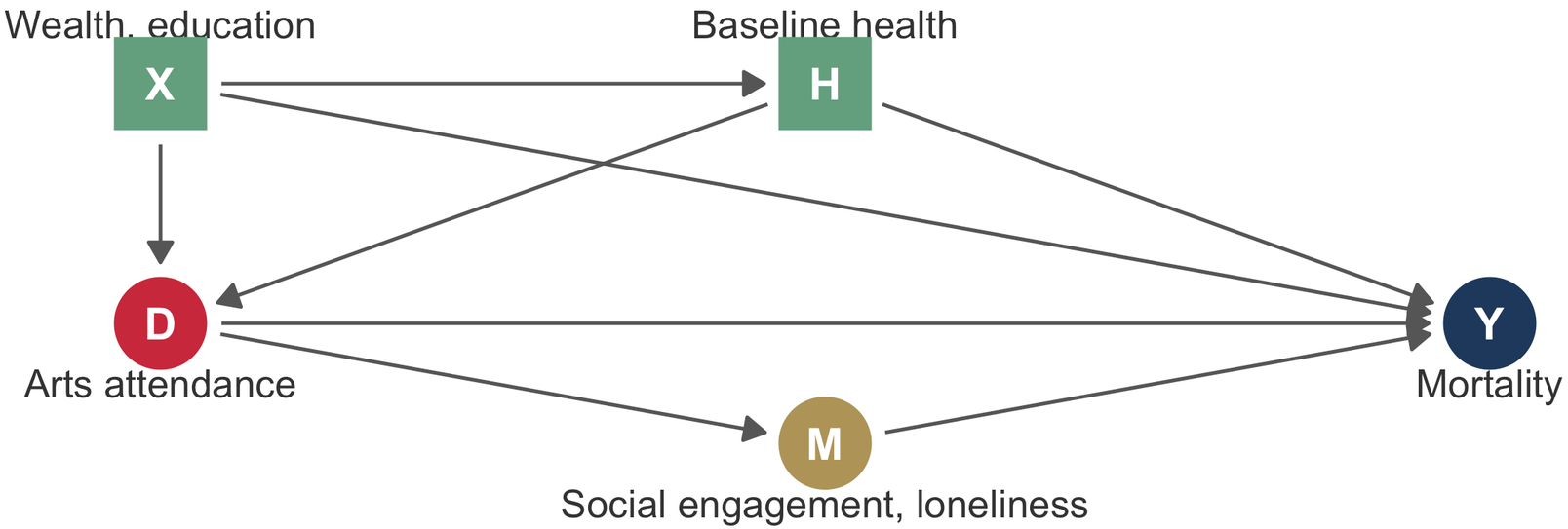
Opera, revisited: suppose this is the DAG

Not *the* true DAG, but **a DAG**: our theory of how arts attendance, health, wealth, and social life fit together, written down.



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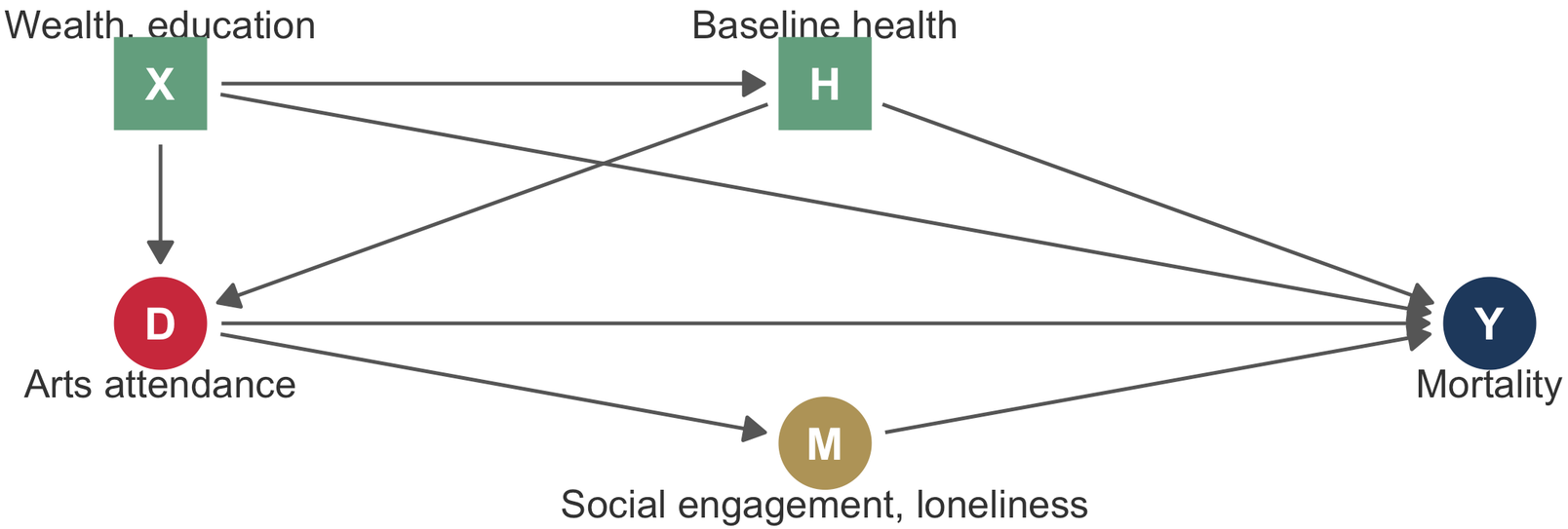
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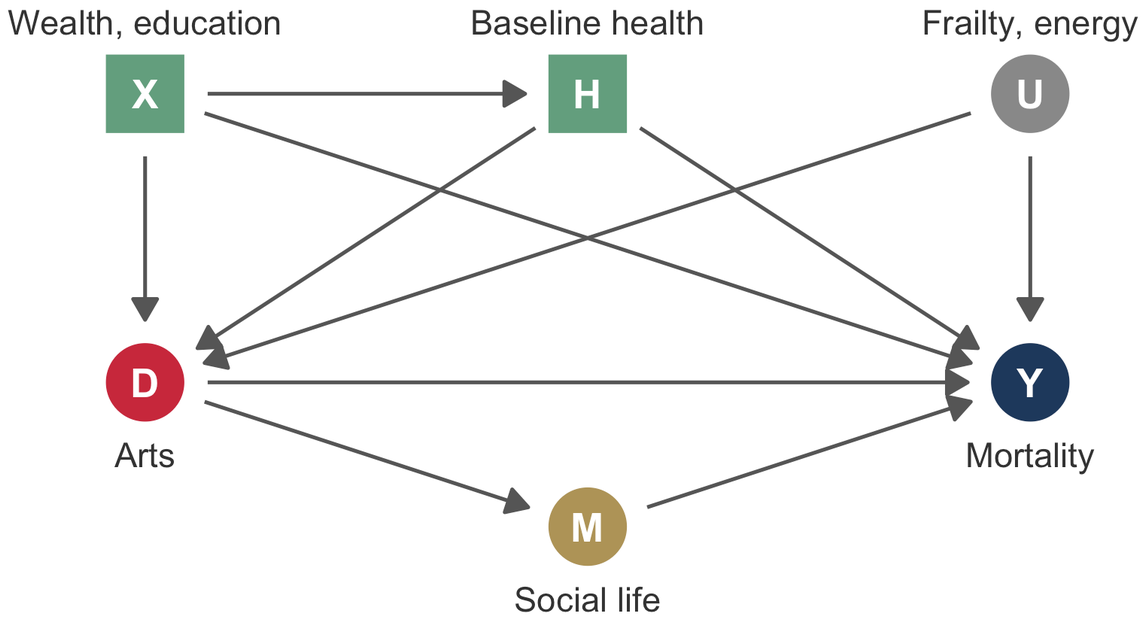
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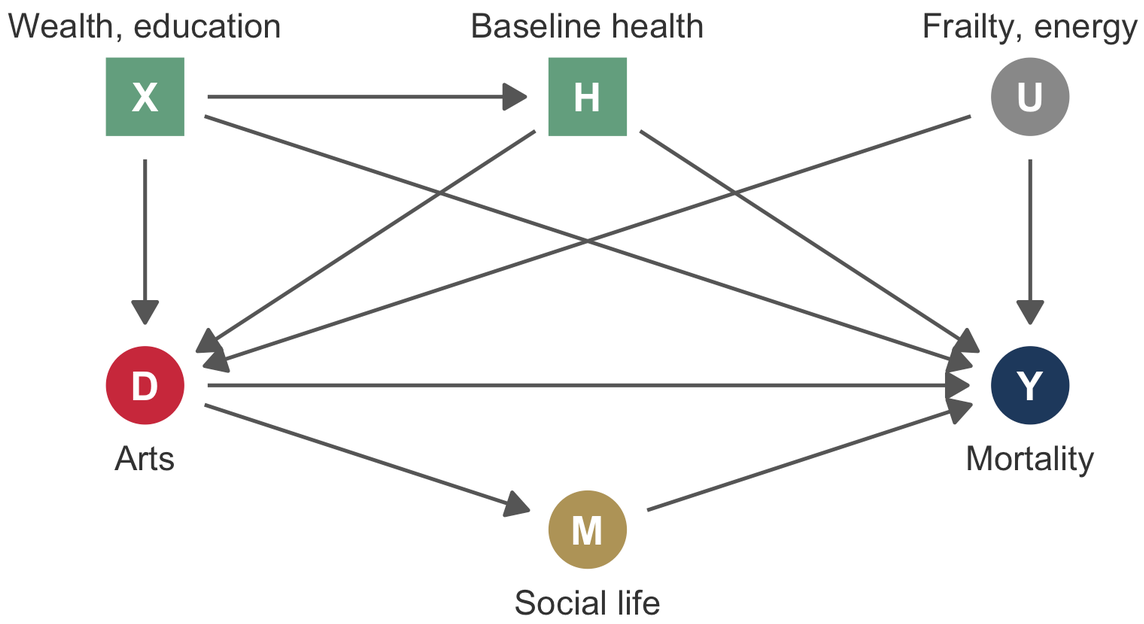
The leverage of a DAG: every arrow, and every **missing** arrow, is a claim **others can see and contest**.

Contest it: a critic’s DAG



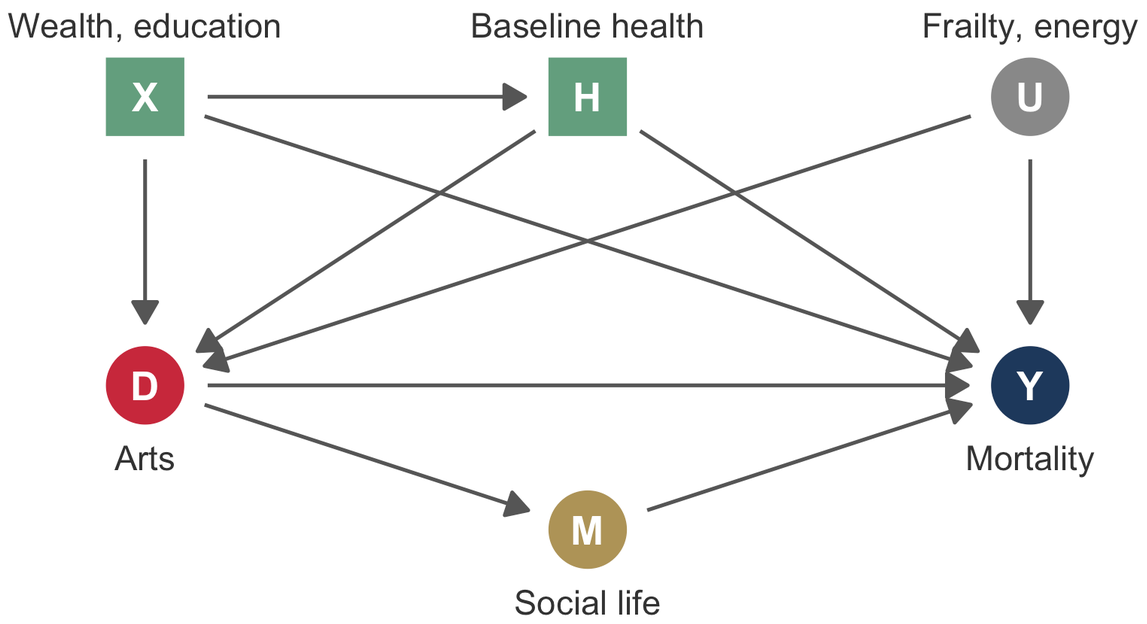
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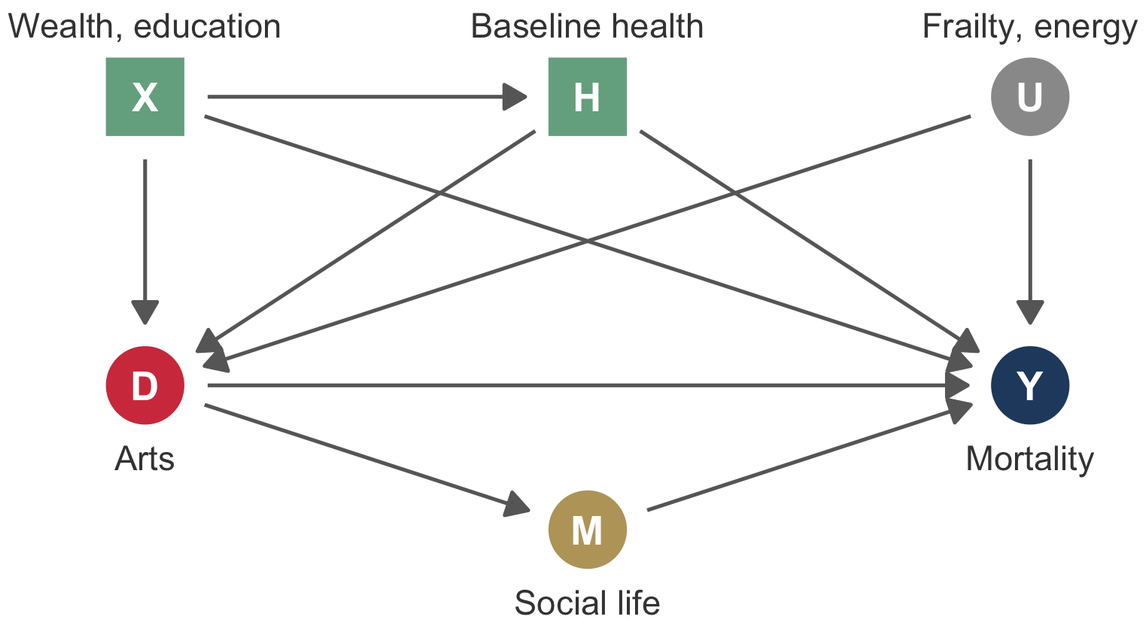
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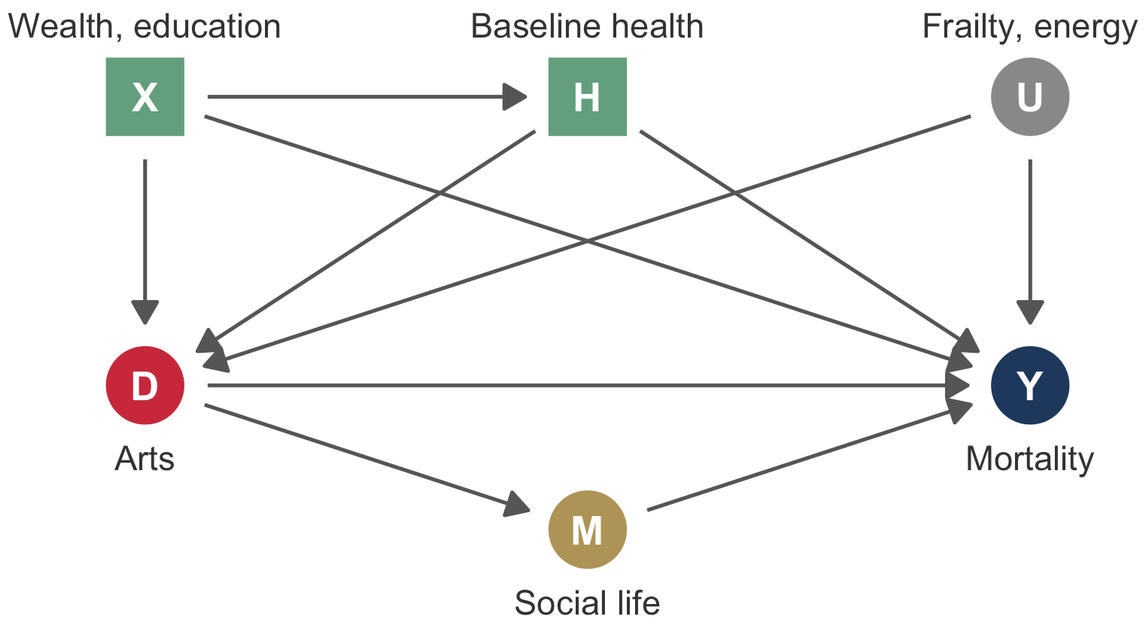
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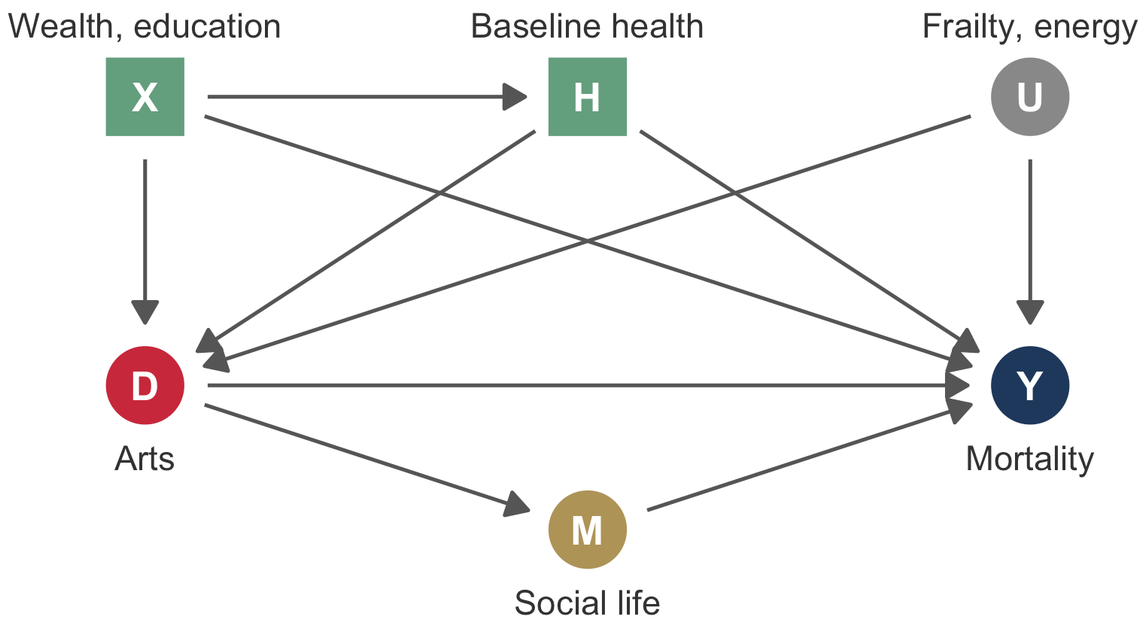
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Disagreeing about a DAG is a **scientific debate, not a matter of taste**: you have to say which arrow, and why.

Where do your data come from?

Is there racial bias in police use of force?

Fryer (2019), "An empirical analysis of racial differences in police use of force", *Journal of Political Economy* 127(3): 1210–1261

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- A question with enormous stakes, and one where **the data are produced by the very behavior under study**. The usual source: **administrative records** of police–civilian encounters.

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- So before judging either number, ask two questions: **what is the estimand, and who is in the data?**

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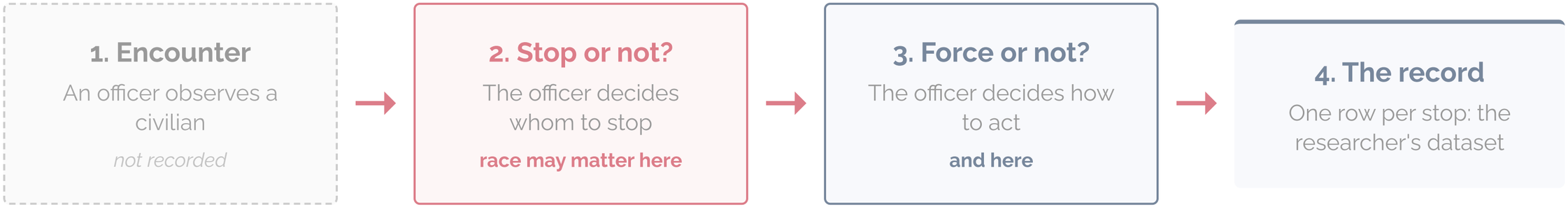
and here

→

4. The record

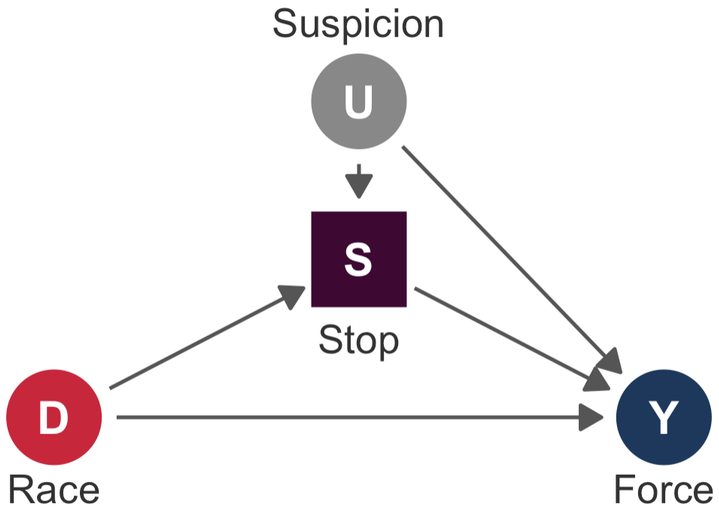
One row per stop: the researcher's dataset

How does a row get into the data?



If race affects step 2, the people in the records are **already a selected group**, before anyone runs a regression.

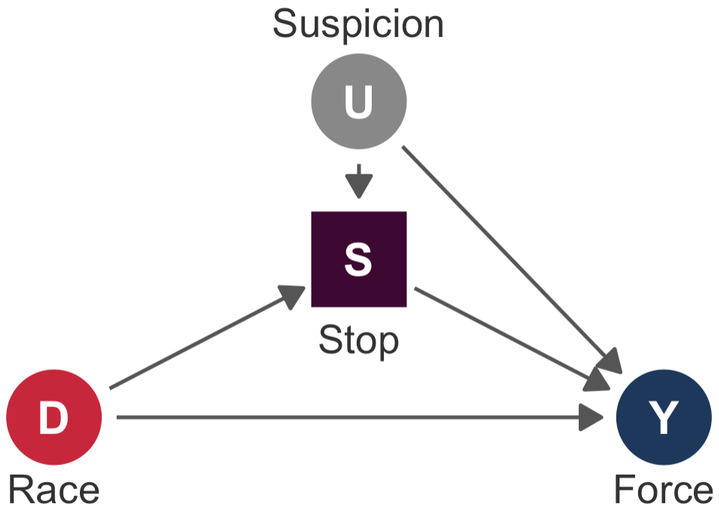
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Knox, Lowe & Mummolo (2020), *APSR* 114(3)

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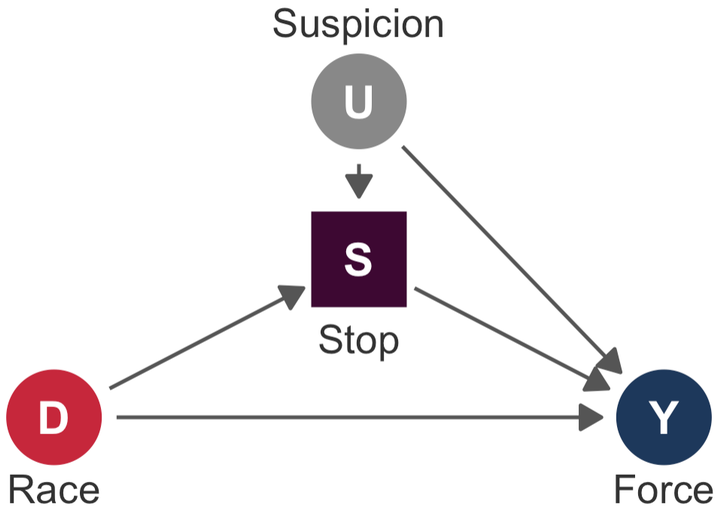
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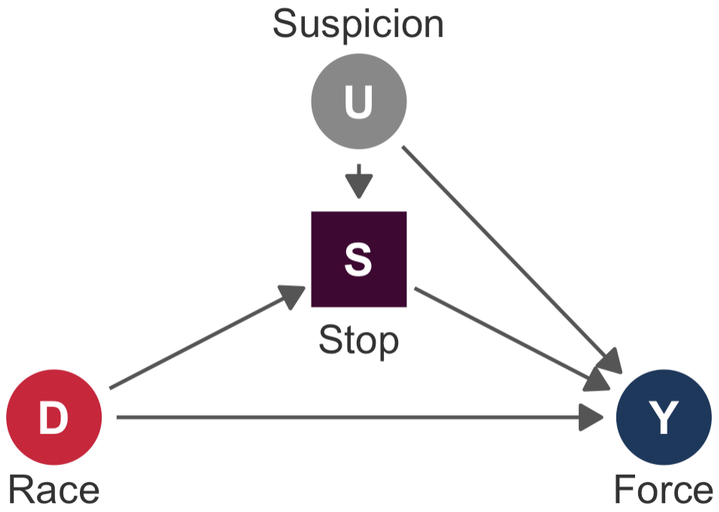
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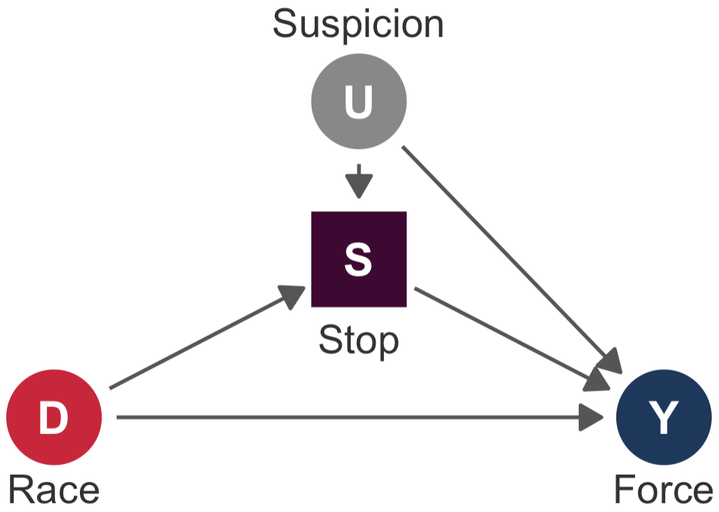
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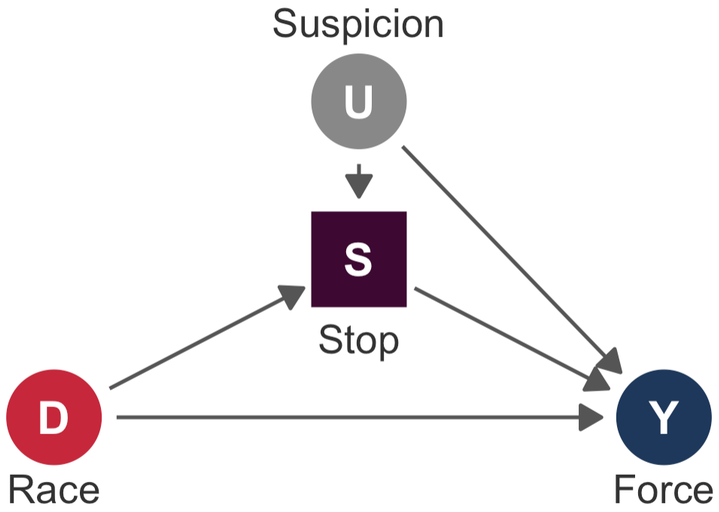
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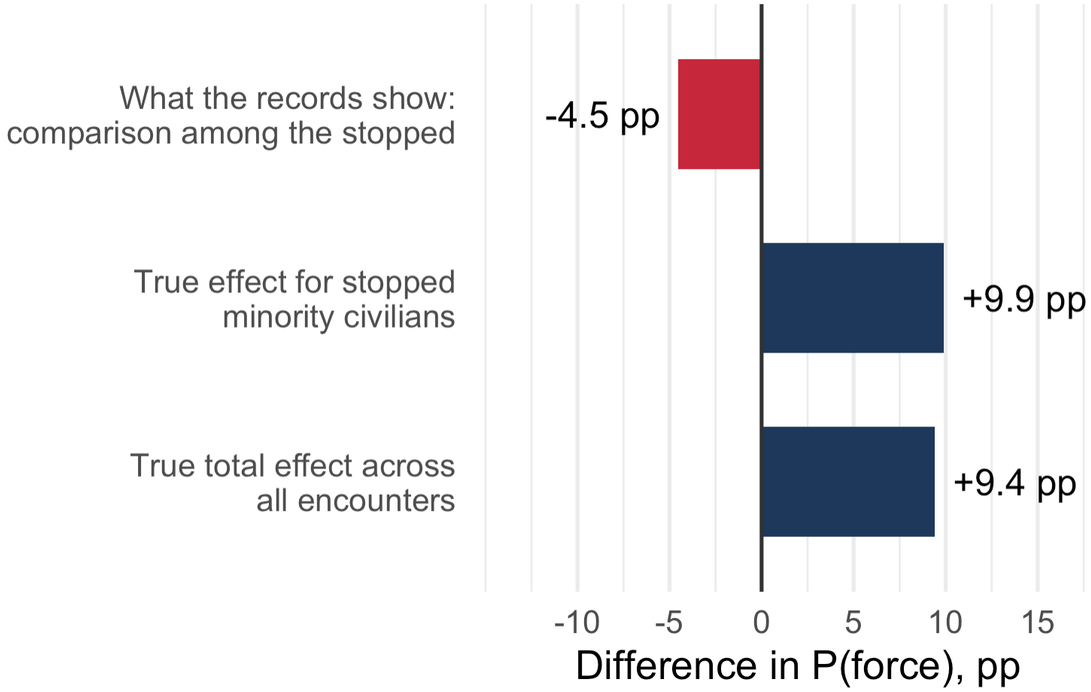
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- And the stop is also a **mediator** ($D \rightarrow S \rightarrow Y$): even the question "what is the effect?" needs care



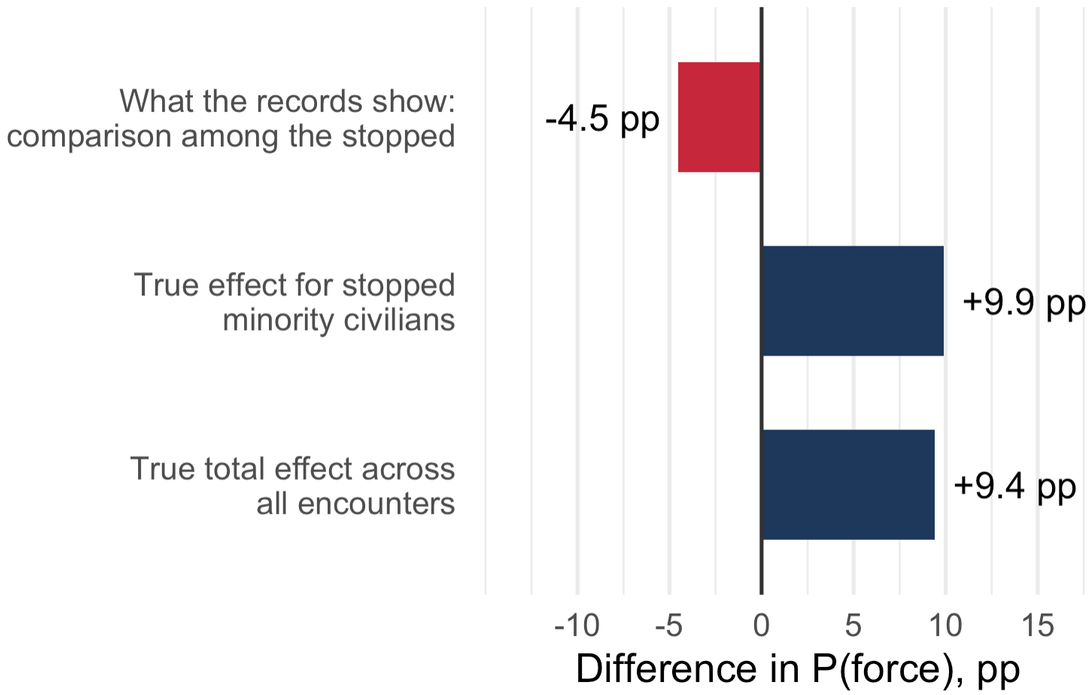
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How selection can hide discrimination



How selection can hide discrimination

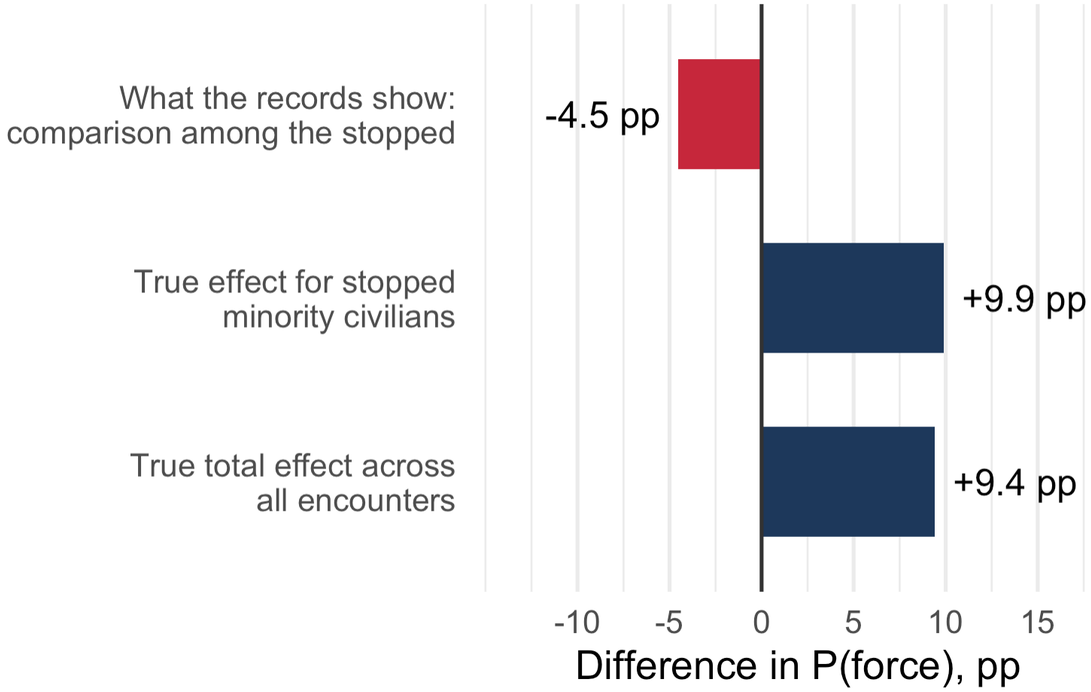
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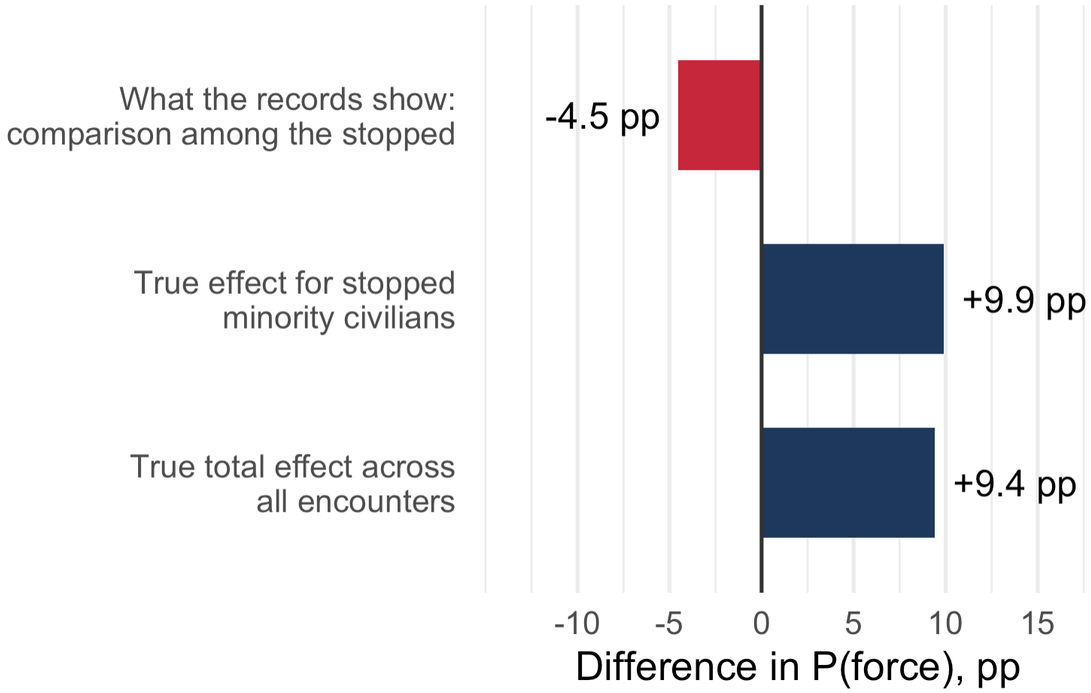
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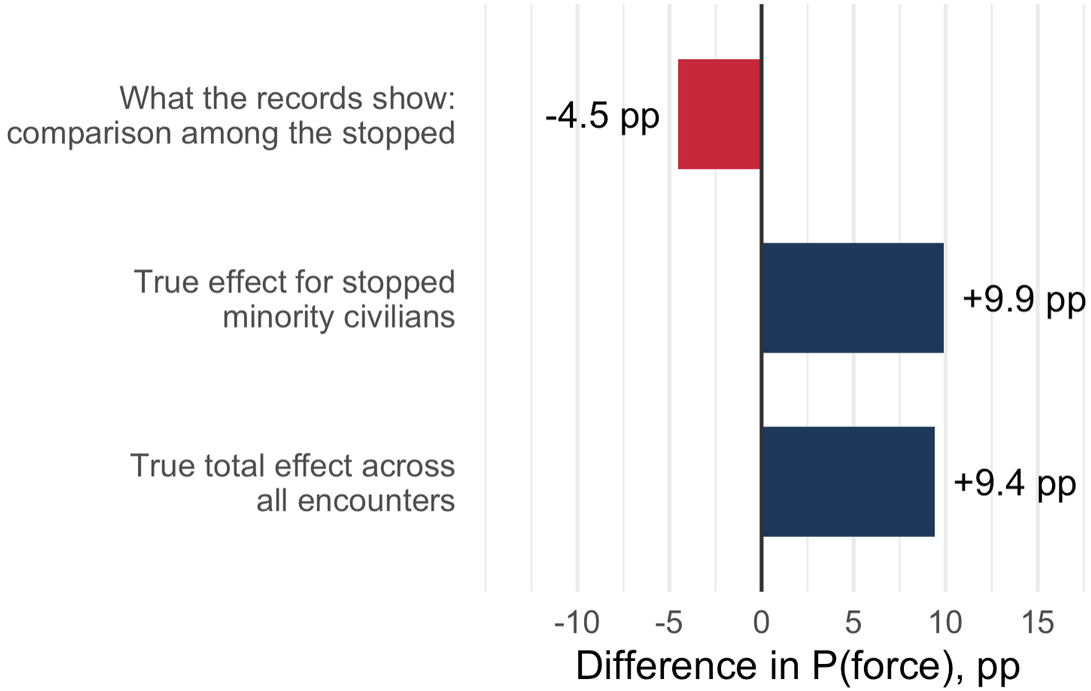
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- In the records, minority civilians face force **4.5 pp less** often
- Yet the true effect is **+9.9 pp** for the same people
- Stopped minority civilians were stopped on **weaker suspicion**, so they look "less dangerous" in the records: the comparison **masks the discrimination entirely**, and a "no difference" finding is no evidence of no bias



Which estimand? Three candidates

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How much does a civilian's race change the chance that an encounter ends in force?

Needs data on encounters that were never recorded

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The data *can probably* answer the middle question; the public (*and policy*) debate ask the first or the third.

Name the estimand first.

What Knox, Lowe & Mummolo do

Knox, Lowe & Mummolo (2020), *APSR* 114(3): 619–637 · Bronner (2020), FiveThirtyEight

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> Bronner (2020) gives the plain-language version: statistics on police interactions describe only the small, non-random subset of people police chose to engage, so the relationships in those data can shift. A collider, in disguise.

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Previous research is part of the data-generating process too. This is a bit of a pet peeve for Kevin and me: published findings are selected on statistical significance (Think of Week 1). The literature you read is a selected sample…

IF TIME ALLOWS

Your turn 💬 (~5 min, groups of 3, then share)

Pick **one** ‘dataset’ (your own, or one below) and draw how a row comes to exist.

1. Asylum decisions in court records
2. Police-reported bicycle crashes (do protected lanes make cycling safer?)
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Where is the selection node? Is it a collider between your treatment and outcome? What estimand do your data actually support?

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LAST SPRING

Advances in Diff-in-Diff

Slides and labs from the spring 2026 workshop: ask me for the materials

Wrapping up

Three questions before you run your regression

> A pre-going crazy with R (or whatever statistical software you use) checklist for any analysis that claims an effect (based on today's content):

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If you cannot answer the first question, the coefficient has no interpretation yet. Choosing controls is a **theoretical** decision, not a statistical one.

What we covered today

FROM QUESTION TO DATA

Causation and counterfactuals

Potential outcomes; ATE, ATT and ATC; the naive comparison = ATE + baseline bias + differential treatment effect bias

Estimands

Unit-specific quantity + target population, stated outside any model; descriptive targets are legitimate too

DAGs and controls

Conditioning on a confounder or mediator closes a path; conditioning on a collider opens one; choose controls from the graph, not from the data

Where data come from

Selection into the data is conditioning; administrative records can mask the effects we care about (Knox, Lowe & Mummolo)

Where this goes next

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WEEK 4 · MEASUREMENT I

We never see Y itself

Today we treated 'outcomes' as numbers in a table. In practice we observe a measure of it, Y*, not Y. What does measurement error do to an estimate, and does a finding travel to other people, places, and times?

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WEEK 5 · MEASUREMENT II

Constructs behind many answers

Sometimes 'outcomes' are not one survey answer but a latent construct behind many. How to measure it with a model, carry its uncertainty forward, and check that it means the same thing in every group we compare.

Readings

Compulsory

- Lundberg, Johnson & Stewart (2021). What is your estimand? *American Sociological Review* 86(3): 532–565.
- Knox, Lowe & Mummolo (2020). Administrative records mask racially biased policing. *APSR* 114(3): 619–637.
- Bronner (2020). Why statistics don't capture the full extent of the systemic bias in policing. *FiveThirtyEight*.

Optional

- Cunningham (2021). *Causal Inference: The Mixtape*, Ch. 3–4.
- Cinelli, Forney & Pearl (2024). A crash course in good and bad controls. *Sociological Methods & Research* 53(3): 1071–1104.
- Elwert & Winship (2014). Endogenous selection bias. *Annual Review of Sociology* 40: 31–53.
- Imbens (2020). Potential outcome and directed acyclic graph approaches to causality. *JEL* 58(4): 1129–1179.

Also cited today

- Behaghel, Crépon & Le Barbanchon (2015). *AEJ: Applied* 7(3): 1–27.
- Boisjoly, Duncan, Kremer, Levy & Eccles (2006). *AER* 96(5): 1890–1905.
- Fancourt & Steptoe (2019). *BMJ* 367: l6377.
- Fryer (2019). *Journal of Political Economy* 127(3): 1210–1261.
- Holland (1986). *JASA* 81(396): 945–960.
- Montgomery, Nyhan & Torres (2018). *AJPS* 62(3): 760–775.
- Sen & Wasow (2016). *Annual Review of Political Science* 19: 499–522.

Your regression returns a number.

Your estimand and your DAG decide

what that number is allowed to mean.

Next week: Measurement I · from concepts to numbers

In the lab on Monday

with Martín

In Lab 3 you'll draw today's DAGs in R, recompute the eight-student table, and simulate what confounders and colliders do to a regression.

`ggplot()` `dagify()` `paths()` `adjustmentSets()` `lm()`

LAB 3

`03-dags-lab.qmd`